

Optimal Transport Bounds for Latent Density Smoothing on MTW($K > 0$) Manifolds

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Abstract

Smoothing an empirical distribution can fill gaps between observations, but noise added in the surrounding space also moves probability mass away from the manifold supporting the data. We study this tradeoff through Wasserstein error bounds for intrinsic heat smoothing, ambient Gaussian smoothing, and smoothing in an encoder–decoder’s latent space. Under the Ma–Trudinger–Wang condition with positive cross-curvature and the stated transport-regularity assumptions, we prove that sufficiently small intrinsic smoothing improves upon the unsmoothed empirical measure. Our ambient bound quantifies how increasing the surrounding dimension restricts the smoothing scale suggested by the bound. For latent smoothing, the analysis shows how the benefit depends on reconstruction accuracy and on how the encoder–decoder pair preserves movement along the manifold and responds to noise in other directions. A quadratic approximation to the bound yields bandwidth-selection rules and conditions favoring latent smoothing in this comparison. Controlled synthetic experiments examine the predicted bandwidth and geometric trends. An exploratory MNIST study illustrates latent smoothing relative to pixel-space perturbations.

Keywords: optimal transport, Wasserstein distance, manifold smoothing, Ma–Trudinger–Wang condition, latent variable models

1 Introduction

High-dimensional data in modern machine learning, such as high-resolution images, biological gene expression profiles, and 3D geometric meshes, typically concentrates near a low-dimensional manifold $M \subset \mathbb{R}^D$ with intrinsic dimension $m \ll D$. A foundational objective in statistical learning is to approximate the unknown target probability distribution μ supported on M using a discrete empirical measure $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ formed by n independent observed samples.

To evaluate estimation fidelity, we employ the 2-Wasserstein distance W_2 . Existing results characterize the convergence of empirical measures in Wasserstein distance and explain how the intrinsic dimensional structure of the target distribution can govern sample complexity (Boissard and Le Gouic, 2014; Weed and Bach, 2019). The precise rates depend on the dimension, distributional assumptions, and probabilistic formulation of the error bound.

Complementary to analyses of convergence as the sample size increases, we study the effect of smoothing at a fixed sample size n . Specifically, we analyze $W_2(k_\sigma * \hat{\mu}_n, \mu)$ relative

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to the unsmoothed manifold-supported target measure μ , with explicit dependence on the Gaussian smoothing bandwidth σ .

Defined in ambient space $\mathbb{R}^D$ as $k_\sigma(x) = (2\pi\sigma^2)^{-D/2} \exp(-\|x\|^2/2\sigma^2)$ for bandwidth $\sigma > 0$, convolving the discrete measure $\hat{\mu}_n$ with k_σ redistributes empirical mass to yield the continuous probability density

$$(k_\sigma * \hat{\mu}_n)(x) = \frac{1}{n} \sum_{i=1}^n k_\sigma(x - x_i). \quad (1)$$

Smoothing spreads mass around each observation and can help fill gaps between samples. When the data lie on a manifold, however, the direction of that movement matters. Movement along the manifold can bring mass closer to unobserved parts of the target distribution, whereas movement away from it introduces distortion. Ambient Gaussian noise produces both effects. As the surrounding dimension increases while the manifold stays fixed, more of the injected noise acts in directions away from the manifold.

We make this tradeoff precise by first studying heat smoothing within the manifold. Under the stated Ma–Trudinger–Wang (MTW), cut-locus, and density assumptions, the regularity of optimal transport potentials allows us to quantify its effect on Wasserstein error (Ma et al., 2005; Figalli, 2010). We prove that sufficiently small positive intrinsic smoothing improves upon the raw empirical measure. We then compare ambient smoothing with this intrinsic reference, separating the contribution of noise away from the manifold from the remaining geometric effects.

The resulting ambient error bound suggests a smaller smoothing bandwidth as the surrounding dimension increases. More precisely, for fixed intrinsic geometry, target distribution, and sample, the bandwidth minimizing its quadratic surrogate scales inversely with ambient dimension; the improvement in that surrogate also shrinks. The detailed coefficients and the restrictions needed to apply this approximation are given in Section 3.

This motivates *latent density smoothing*: encode the observations in a lower-dimensional space, add Gaussian noise there, and decode the perturbed points. Reducing the noise dimension can help, but the encoder–decoder geometry also matters. A useful representation must reconstruct the data accurately, preserve movement along the manifold, and control the decoder’s response to noise in other latent directions. Our latent bound quantifies these effects and gives sufficient conditions for its optimized quadratic surrogate to be smaller than the ambient counterpart.

The comparison concerns quadratic surrogates of upper bounds, rather than the exact Wasserstein errors. Applying it to the complete bounds requires controlling the remainder terms. Moreover, the common reference is the intrinsic empirical error, which can exceed the ambient empirical error. The numerical experiments in Section 5 examine whether the predicted trends also appear in the computed transport errors.

1.1 Summary of Contributions

Our main contributions are as follows:

- We derive a small-bandwidth ambient Wasserstein error bound that separates smoothing within the manifold from noise away from it and additional geometric corrections (Theorem 1).

- Under the stated geometric and transport-regularity assumptions, we prove that sufficiently small positive intrinsic heat smoothing improves the squared intrinsic Wasserstein error of the empirical measure (Theorem 2).
- We derive a latent smoothing bound that accounts for reconstruction accuracy and the local geometry of the encoder–decoder pair. We obtain bandwidth-selection rules from its quadratic surrogate and conditions under which the optimized latent surrogate is smaller than its ambient counterpart (Theorem 3).
- We test the predicted bandwidth and geometric trends on controlled synthetic examples, and compare latent perturbations with a pixel-space baseline in an exploratory MNIST study (Section 5).

1.2 Related Work

Our work intersects statistical measure estimation on submanifolds, kernel regularization under optimal transport, score-based generative modeling, and geometric representation learning.

Statistical Estimation on Submanifolds. An extensive literature studies empirical and nonparametric measure estimation under Wasserstein loss (Dudley, 1969; Fournier and Guillin, 2015; Weed and Bach, 2019; Niles-Weed and Berthet, 2022). When the target is supported on or near a smooth low-dimensional manifold, convergence rates can depend on the intrinsic geometry and density regularity rather than directly on the ambient dimension. In particular, Divol (2022) constructs kernel-based estimators for manifold-supported measures and establishes intrinsic-dimensional minimax rates. Our analysis addresses a different question: for a fixed empirical measure, we quantify the local dependence of intrinsic, ambient, and encoder–decoder smoothing errors on the bandwidth and representation geometry.

Gaussian-Smoothed Optimal Transport. Building on classical non-parametric density estimation (Rosenblatt, 1956; Parzen, 1962), recent studies analyze the statistical convergence of smoothed measures under optimal transport (Goldfeld et al., 2020; Nietert et al., 2021; Goldfeld et al., 2024). For a fixed bandwidth $\sigma > 0$, Gaussian convolution achieves a parametric fluctuation rate $W_2(k_\sigma * \hat{\mu}_n, k_\sigma * \mu) = \mathcal{O}(n^{-1/2})$, though the approximation bias grows with the ambient dimension D (Goldfeld et al., 2020). We extend this line of work by characterizing how smoothing bandwidth dictates the W_2^2 bias-variance trade-off specifically on low-dimensional submanifolds.

Score-Based Diffusion Models. Score-based diffusion models directly leverage ambient Gaussian smoothing by constructing a continuous family of convolved measures $k_\sigma * \hat{\mu}_n$ across noise scales $\sigma > 0$ (Song et al., 2020; Ho et al., 2020). In this paradigm, the forward process corresponds to progressive ambient kernel convolution, while the reverse process learns time-dependent score functions $\nabla_x \log(k_\sigma * \mu)(x)$ to invert the smoothing transformation as $\sigma \rightarrow 0$ (Hyvärinen and Dayan, 2005; Vincent, 2011). Theoretical studies demonstrate that these reverse dynamics can adapt to low-dimensional manifold structures $\mathcal{M} \subset \mathbb{R}^D$, as the learned vector fields act as restoring drift terms that contract off-manifold probability mass

along the $D - m$ normal directions of $\mathcal{M}$ (Pidstrigach, 2022; De Bortoli, 2022; Oko et al., 2023).

To mitigate the computational cost of high-dimensional ambient score matching, Latent Diffusion Models (LDMs) apply this noisy forward-reverse mechanism within a learned latent space (Rombach et al., 2022). While LDMs treat the latent space primarily as an empirical computational shortcut, our work rigorously characterizes the non-asymptotic W_2 transport error under latent smoothing, formalizing the geometric trade-offs between latent bandwidth selection and representation distortion.

Intrinsic vs. Extrinsic Manifold Smoothing. Smoothing probability measures on submanifolds can be formulated intrinsically via Riemannian heat kernels k_σ^M (Bérard et al., 1994; Coifman and Lafon, 2006) or extrinsically via ambient kernel convolution in $\mathbb{R}^D$ (Niyogi et al., 2008). Intrinsic heat kernels preserve underlying manifold geometry but require explicit access to the metric tensor, whereas extrinsic ambient convolution is computationally direct but diffuses mass off-manifold. We bridge these perspectives by using the geometric reach (Federer, 1959) to bound the optimal transport discrepancy between extrinsic ambient convolution and intrinsic Riemannian smoothing.

Geometric Regularization in Deep Representations. Mapping high-dimensional data into low-dimensional coordinate spaces while controlling geometric distortion is a foundational objective in manifold learning (Tenenbaum et al., 2000; Belkin and Niyogi, 2003). Modern deep representation models rely on architectural constraints, such as spectral normalization (Miyato et al., 2018), and denoising objectives (Vincent et al., 2008; Alain and Bengio, 2014), to regulate representation geometry. Our work complements this literature by providing theoretical bounds that directly link Jacobian-based geometric properties of encoder-decoder mappings to optimal transport performance under latent smoothing.

1.3 Organization of the Paper.

The remainder of this paper is organized as follows. Section 2 introduces the formal problem setup, optimal transport prerequisites, and geometric foundations under the MTW($K > 0$) condition. Section 3 states our main theoretical results, Theorems 1 to 3, and provides detailed interpretations of the error decompositions, optimal bandwidths, and estimation gains. Section 4 presents the complete proofs for all main theoretical bounds. Finally, Section 5 validates our theoretical framework through numerical experiments, structured into three parts: empirical validation of ambient smoothing bounds (Theorems 1 and 2), evaluation of latent space density smoothing performance (Theorem 3), and application to real-world image distributions using MNIST.

2 Problem Setup and Geometric Foundations

We formalize the geometric, measure-theoretic, and architectural setup used throughout the paper.

Submanifold Geometry and Reach Let $M \subset \mathbb{R}^D$ be a smooth, compact, connected m -dimensional Riemannian manifold embedded in ambient euclidean space $\mathbb{R}^D$, where $m \ll D$.

Definition 1 (Reach). *The reach of M , denoted by $\tau = \text{reach}(M) > 0$, is the supremum over all $r > 0$ such that every point $y \in \mathbb{R}^D$ within an open r -neighborhood $U_r(M) = \{y \in \mathbb{R}^D : \text{dist}(y, M) < r\}$ has a unique metric projection $\pi_M(y) \in M$.*

For any $x \in M$, the ambient space decomposes into the orthogonal sum $\mathbb{R}^D = T_x M \oplus N_x M$, where $T_x M \cong \mathbb{R}^m$ is the tangent space and $N_x M \cong \mathbb{R}^{D-m}$ is the normal space. The reach τ bounds the local curvature of M : for any $x, y \in M$, the normal displacement satisfies $\|\pi_{N_x M}(y - x)\| \leq \frac{\|y - x\|^2}{2\tau}$.

Noise Model and Measures. Let μ be an unknown smooth, strictly positive probability density supported on M satisfying

$$0 < \rho_{\min} \leq \mu(x) \leq \rho_{\max} < \infty \quad \text{for every } x \in M.$$

In practice, we observe an empirical sample $X_n = \{x_1, \dots, x_n\}$, defining the clean empirical measure $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. To model realistic observational corruption, we assume the observed data is corrupted by isotropic ambient noise of scale $\sigma > 0$. This yields the extrinsically smoothed empirical measure

$$\mu_{\text{amb}} = k_\sigma * \hat{\mu}_n, \quad \text{where } k_\sigma(y) = (2\pi\sigma^2)^{-D/2} \exp\left(-\frac{\|y\|^2}{2\sigma^2}\right). \quad (2)$$

Note that we use k_σ to denote the isotropic Gaussian kernel operating in either the ambient space $\mathbb{R}^D$ or the latent space $\mathbb{R}^d$; despite this slight abuse of notation, the intended dimension will be clear from context throughout. Conversely, the ideal intrinsically smoothed measure on M is defined via the Riemannian heat kernel $k_\sigma^M(x, y)$

$$\mu_\sigma = k_\sigma^M * \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n k_\sigma^M(x_i, \cdot). \quad (3)$$

Wasserstein distance convention. We write W_2 for the 2-Wasserstein distance associated with the ambient Euclidean cost

$$c_{\text{amb}}(x, y) = \|x - y\|^2,$$

whereas $W_{2,M}$ denotes the intrinsic 2-Wasserstein distance on M associated with the squared geodesic cost

$$c_M(x, y) = d_M^2(x, y).$$

Since $\|x - y\| \leq d_M(x, y)$, $x, y \in M$, any two probability measures ν_1, ν_2 supported on M satisfy

$$W_2(\nu_1, \nu_2) \leq W_{2,M}(\nu_1, \nu_2). \quad (4)$$

We use $W_{2,M}$ to analyze intrinsic heat smoothing on M and W_2 whenever at least one of the measures may be supported outside M , as occurs under ambient or latent-space smoothing.

3 Main Theoretical Bounds

To establish the statistical properties of smoothed empirical estimators when probability mass concentrates on lower-dimensional structures, consider an unknown target probability distribution μ supported on an m -dimensional compact Riemannian manifold $M \subset \mathbb{R}^D$ with reach $\tau > 0$. Given a finite empirical sample $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ drawn independently from μ , raw discrete empirical estimators suffer from severe finite-sample variance. While injecting noise provides regularization by smoothing out discrete empirical fluctuations, it forces probability mass to interact with the global geometry of the embedding space.

The widely used approach to continuous density estimation is ambient Gaussian smoothing, wherein the empirical distribution is convolved directly with an isotropic Gaussian kernel k_σ in the embedding space $\mathbb{R}^D$. However, because the ambient dimension is typically far larger than the manifold dimension ($D \gg m$), ambient isotropic noise disperses probability mass into normal directions orthogonal to the tangent bundle of M . By decomposing the overall Wasserstein error into an intrinsic manifold error and an off-manifold projection penalty, we establish the following non-asymptotic upper bound.

Theorem 1 (Ambient Smoothing Error Bound). *Let $M \subset \mathbb{R}^D$ be a compact m -dimensional submanifold with reach $\tau > 0$, and let μ be a smooth probability measure supported on M . Let $\mu_\sigma = k_\sigma^M * \hat{\mu}_n$ denote the intrinsic heat-smoothed empirical measure at diffusion time $t = \sigma^2$. For sufficiently small bandwidth $\sigma > 0$, the ambiently smoothed empirical measure $k_\sigma * \hat{\mu}_n$ satisfies*

$$W_2^2(k_\sigma * \hat{\mu}_n, \mu) \leq W_2^2(\mu_\sigma, \mu) + \sigma^2(D - m) + \frac{4mW_2(\hat{\mu}_n, \mu)\sigma^2}{\tau} + \mathcal{O}\left(\frac{\sigma^3}{\tau}\right) + C_{\text{tail}}, \quad (5)$$

where the constant implicit in $\mathcal{O}(\sigma^3/\tau)$ is independent of σ but may depend on m , D , and the geometry of M . The nonnegative tail term $C_{\text{tail}} = C_{\text{tail}}(\tau, \sigma, m, D, M)$ is specified in (31), under the restriction $0 < \sigma < \tau/\sqrt{mD}$. For fixed m , D , and M , this term decays exponentially in $1/\sigma^2$ as $\sigma \rightarrow 0$.

Theorem 1 isolates the normal-noise contribution $\sigma^2(D - m)$ relative to intrinsic smoothing. It is the expected squared displacement in the $D - m$ normal directions. After applying Theorem 2, this contribution combines with $m\sigma^2$ to give $D\sigma^2$, in addition to the geometric correction. Thus codimension describes the normal component, whereas ambient dimension governs the explicit dimensional term in the combined bound.

To determine the statistical benefit of smoothing, one must analyze the intrinsic error term $W_{2,M}(\mu_\sigma, \mu)$, which captures finite-sample variance reduction along the Riemannian heat flow $\partial_t \mu_\sigma = \frac{1}{2} \Delta_M \mu_\sigma$ on M for $t = \sigma^2$. Quantifying the exact rate at which heat diffusion smoothes discrete empirical spikes requires controlling the temporal derivative of the intrinsic optimal transport cost. This derivative depends on the spatial regularity of optimal transport maps, which is guaranteed under the Ma–Trudinger–Wang (MTW) cross-curvature condition (Ma et al., 2005; Figalli, 2010).

Definition 2 (The MTW(K) Condition). *Let (M, g) be a Riemannian manifold, and let $c(x, y) = \frac{1}{2}d_M^2(x, y)$ denote the squared intrinsic geodesic cost. For non-focal pairs $x, y \in M$, let $p = -\nabla_x c(x, y) \in T_x M$ be the initial velocity vector of the geodesic from x to y . The*

Ma–Trudinger–Wang (MTW) tensor $\mathfrak{S}_{(x,y)}(\xi, \eta)$ measures the fourth-order cross-derivative of $c(x, y)$ under orthogonal perturbations $\xi, \eta \in T_x M$ with $\langle \xi, \eta \rangle_g = 0$. The manifold satisfies the MTW(K) condition for $K \in \mathbb{R}$ if

$$\mathfrak{S}_{(x,y)}(\xi, \eta) \geq K \|\xi\|_g^2 \|\eta\|_g^2.$$

Remark 1 (Geometric Intuition and Transport Regularity). When $K > 0$, the manifold satisfies the strict MTW condition, which imposes a positive lower bound on the cross-curvature of the transport cost. Geometrically, this requires M to exhibit positive curvature characteristics (such as round spheres or complex projective spaces) rather than negative, saddle-like curvature. Crucially, as established by [Figalli \(2010\)](#), strict MTW curvature combined with a non-focal cut-locus guarantees global C^∞ smoothness of optimal transport maps and their dual Kantorovich potentials.

Equipped with the global C^∞ regularity of dual transport potentials, differentiating the squared intrinsic Wasserstein distance $W_{2,M}^2(\mu_\sigma, \mu)$ along the Riemannian heat flow trajectory $\partial_t \mu_\sigma = \frac{1}{2} \Delta_M \mu_\sigma$ for $t = \sigma^2$ yields an explicit non-asymptotic variance reduction bound.

Theorem 2 (Intrinsic Variance Reduction Rate). Let (M, g) be a compact, connected, m -dimensional Riemannian manifold without boundary. Assume that M has a nonfocal cut locus and that the squared geodesic cost $c(x, y) = \frac{1}{2} d_M^2(x, y)$ satisfies the MTW(K) condition for some $K > 0$. Let $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ be the empirical measure of independent samples $x_i \sim \mu$, and let $\mu_\sigma = k_\sigma^M * \hat{\mu}_n$ denote its intrinsic heat smoothing at time $t = \sigma^2$, with heat generator $\frac{1}{2} \Delta_M$. For all sufficiently small $\sigma > 0$, the squared intrinsic Wasserstein distance satisfies

$$W_{2,M}^2(\mu_\sigma, \mu) \leq W_{2,M}^2(\hat{\mu}_n, \mu) - C_m n^{-1/m} \sigma + m \sigma^2, \quad (6)$$

where $C_m = \frac{m \rho_{\max}^{-1/m}}{\sqrt{2\pi}} \left(\frac{m}{m+1} \right)^{m/2}$.

Theorem 2 gives an explicit error reduction through the leading $-C_m n^{-1/m} \sigma$ term. Its geometric assumptions restrict its scope. In particular, the MTW(K) condition is known mainly for round spheres, complex projective spaces, and their small perturbations, excluding many Riemannian manifolds of interest.

Extending this intrinsic derivative framework to general Riemannian manifolds without the MTW condition remains a challenging open problem. A natural alternative is the partial regularity theory of [De Philippis and Figalli \(2015\)](#) (Theorem 1.4), which establishes that for $C^{k,\alpha}$ regular densities, the optimal transport map is a $C_{\text{loc}}^{k+1,\alpha}$ diffeomorphism outside of closed singular sets $\Sigma_X, \Sigma_Y \subset M$ of measure zero.

Applying partial regularity to the present argument requires additional justification. Our proof differentiates $W_{2,M}^2(\mu_{\sqrt{t}}, \mu)$ with respect to diffusion time $t = \sigma^2$ and uses integration by parts for the transport potential. Smoothness outside a set of volume measure zero does not by itself justify this global integration-by-parts step: possible singular contributions to the distributional Laplacian must also be controlled. Moreover, partial regularity alone does not supply the pointwise Laplacian estimate used in our proof. Extending this argument beyond the stated geometric assumptions therefore requires further analysis.

To reduce the ambient-dimensional contribution to the combined bound, modern generative models perform smoothing within a lower-dimensional latent space $\mathbb{R}^d$, where $m \leq d \ll D$. Mass is smoothed in latent space via an encoder $f : \mathbb{R}^D \rightarrow \mathbb{R}^d$ and mapped back to ambient space through a decoder $g : \mathbb{R}^d \rightarrow \mathbb{R}^D$.

To characterize the fidelity of the autoencoder pair (f, g) near data points $x_i \in M$, we introduce three geometric parameters:

- **Reconstruction Error (δ):** The root-mean-square reconstruction error

$$\delta = \left(\frac{1}{n} \sum_{i=1}^n \|x_i - g(f(x_i))\|^2 \right)^{1/2}, \quad (7)$$

quantifying how accurately the autoencoder preserves empirical data points on average.

- **Tangential Gain ($\rho_{\parallel}$) and Isometry Defect (ε_{iso}):** Defined via

$$\varepsilon_{\text{iso}} = \max_{1 \leq i \leq n} \min_{O \in O(T_{x_i}M)} \|A_i - \rho_{\parallel} \iota_i O\|, \quad (8)$$

where $A_i := J_g(f(x_i))J_f(x_i)|_{T_{x_i}M} : T_{x_i}M \rightarrow \mathbb{R}^D$, $O(T_{x_i}M)$ is the orthogonal group of the tangent space, and $\iota_i : T_{x_i}M \hookrightarrow \mathbb{R}^D$ is the inclusion map. The parameter $\rho_{\parallel} > 0$ represents the global tangential scaling factor along the manifold, while $\varepsilon_{\text{iso}} \geq 0$ measures the worst-case local departure of the encoder-decoder composition from a conformal tangential isometry.

- **Orthogonal Expansion Factor (L_{orth}):** Defined via $\|J_g(f(x_i))v^{\perp}\| \leq L_{\text{orth}}\|v^{\perp}\|$ for normal vectors $v^{\perp} \in V_i^{\perp}$ orthogonal to the latent tangent subspace $V_i = J_f(x_i)(T_{x_i}M)$. This factor measures how much the decoder amplifies noise added in directions perpendicular to the manifold representation in latent space.

Theorem 3 (Latent-Space Smoothing Error Bound). *Let $M \subset \mathbb{R}^D$ be a compact, connected m -dimensional submanifold with reach $\tau > 0$. Assume the hypotheses of Theorem 2, or the round-sphere setting of Remark 2, so that the intrinsic smoothing bound applies. Let μ be a smooth probability density supported on M , and let $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ be an empirical measure. Let $f : \mathbb{R}^D \rightarrow \mathbb{R}^d$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}^D$, where $m \leq d \leq D$. For every empirical point x_i , assume that $U_i := J_f(x_i)|_{T_{x_i}M} : T_{x_i}M \rightarrow V_i$ is a linear isometry onto an m -dimensional subspace $V_i \subset \mathbb{R}^d$. Assume also that $g \in C^2(\mathbb{R}^d; \mathbb{R}^D)$ has uniformly bounded first and second derivatives. For $A_i := J_g(f(x_i))J_f(x_i)|_{T_{x_i}M}$, assume that there is one global tangential gain $\rho_{\parallel} > 0$ and define*

$$\varepsilon_{\text{iso}} := \max_{1 \leq i \leq n} \min_{O \in O(T_{x_i}M)} \|A_i - \rho_{\parallel} \iota_i O\|_{\text{op}},$$

where $\iota_i : T_{x_i}M \hookrightarrow \mathbb{R}^D$ is the inclusion. Assume also that

$$\|J_g(f(x_i))v^{\perp}\| \leq L_{\text{orth}}\|v^{\perp}\|, \quad v^{\perp} \in V_i^{\perp}.$$

Set $W_0 := W_{2,M}(\hat{\mu}_n, \mu)$. For sufficiently small $\sigma > 0$ such that $\rho_{\parallel}\sigma$ lies in the small-bandwidth regime of Theorem 2 and $0 < \sigma < \frac{1}{2\sqrt{m}\rho_{\parallel}} \min\{W_0, 2\tau\}$, the generated pushforward measure $\mu_{\text{out}} = g_{\#}((f_{\#}\hat{\mu}_n) * \mathcal{N}(0, \sigma^2 \mathbf{I}_d))$ satisfies

$$W_2^2(\mu_{\text{out}}, \mu) \leq \mathcal{E}_{\text{const}} - \Gamma_{\text{eff}}\sigma + L_{\text{latent}}\sigma^2 + \mathcal{O}(\sigma^3), \quad (9)$$

where

- $\mathcal{E}_{\text{const}} = (1 + \varepsilon_{\text{iso}})W_0^2 + 2\delta W_0 + \delta^2$ is the constant term in the derived upper bound. The contribution $\varepsilon_{\text{iso}}W_0^2$ is introduced by the Young inequality used to control the tangential-defect cross term and should not be interpreted as part of the exact zero-noise reconstruction error,
- $\Gamma_{\text{eff}} = \rho_{\parallel}C_m n^{-1/m} \left(1 + \frac{\delta}{W_0}\right)$ is the effective tangential variance reduction rate, driven by global tangential gain $\rho_{\parallel}$ and augmented by the reconstruction ratio $\frac{\delta}{W_0}$, and $C_m > 0$ is the intrinsic rate constant,
- $L_{\text{latent}} = L_{\text{orth}}^2(d-m) + \varepsilon_{\text{iso}}^2 m + Q$ is the second-order latent noise dispersion coefficient, combining off-manifold noise leakage across codimension $d-m$ with metric distortion and manifold geometric bounds Q .

Theorem 3 shows that latent smoothing replaces the ambient-dimensional noise penalty with a second-order coefficient governed by the latent geometry. In particular, write

$$\beta_{\text{lat}} := L_{\text{latent}} = L_{\text{orth}}^2(d-m) + \varepsilon_{\text{iso}}^2 m + Q.$$

Here Q includes the intrinsic tangential contribution and reconstruction and geometric corrections; β_{lat} is therefore not simply $L_{\text{orth}}^2(d-m)$. Combining Theorems 1, 2 and 3 gives the small-bandwidth upper bounds

$$W_2^2(k_{\sigma} * \hat{\mu}_n, \mu) \leq \Psi_{\text{amb}}(\sigma) + \mathcal{R}_{\text{amb}}(\sigma), \quad (10)$$

$$W_2^2(\mu_{\text{out}}(\sigma), \mu) \leq \Psi_{\text{lat}}(\sigma) + \mathcal{R}_{\text{lat}}(\sigma), \quad (11)$$

where

$$\Psi_{\text{amb}}(\sigma) := W_0^2 - \alpha_n \sigma + \beta_{\text{amb}} \sigma^2, \quad (12)$$

$$\Psi_{\text{lat}}(\sigma) := \mathcal{E}_{\text{const}} - \Gamma_{\text{eff}} \sigma + \beta_{\text{lat}} \sigma^2, \quad (13)$$

and

$$\mathcal{R}_{\text{amb}}(\sigma) = \mathcal{O}(\sigma^3) + C_{\text{tail}}(\tau, \sigma, m, D, M), \quad \mathcal{R}_{\text{lat}}(\sigma) = \mathcal{O}(\sigma^3).$$

Here

$$\alpha_n := C_m n^{-1/m}, \quad \Gamma_{\text{eff}} := \rho_{\parallel} \alpha_n \left(1 + \frac{\delta}{W_0}\right),$$

and we take

$$\beta_{\text{amb}} := D + \frac{4m}{\tau} W_2(\hat{\mu}_n, \mu).$$

Indeed, the intrinsic estimate contributes $m\sigma^2$, and the normal-noise term contributes $(D-m)\sigma^2$, whose sum is $D\sigma^2$, and the geometric cross term contributes $4mW_2(\hat{\mu}_n, \mu)\sigma^2/\tau$. For fixed intrinsic geometry, sample, and target measure, under an isometric inclusion into a larger ambient space, β_{amb} therefore grows linearly with D . The notation $\mu_{\text{out}}(\sigma)$ emphasizes the dependence of the generated measure on the latent bandwidth.

The functions Ψ_{amb} and Ψ_{lat} are the quadratic, second-order surrogates obtained by omitting the remainder terms from (10) and (11). Assuming $\beta_{\text{amb}} > 0$ and $\beta_{\text{lat}} > 0$, their unconstrained minimizers over $\sigma \geq 0$ are

$$\sigma_{\text{amb}}^* := \arg \min_{\sigma \geq 0} \Psi_{\text{amb}}(\sigma) = \frac{\alpha_n}{2\beta_{\text{amb}}}, \quad (14)$$

$$\sigma_{\text{lat}}^* := \arg \min_{\sigma \geq 0} \Psi_{\text{lat}}(\sigma) = \frac{\Gamma_{\text{eff}}}{2\beta_{\text{lat}}}. \quad (15)$$

These expressions are relevant when the resulting bandwidths lie inside the small-bandwidth regimes of the corresponding theorems. The minimum surrogate values are

$$\min_{\sigma \geq 0} \Psi_{\text{amb}}(\sigma) = W_0^2 - \frac{\alpha_n^2}{4\beta_{\text{amb}}}, \quad (16)$$

$$\min_{\sigma \geq 0} \Psi_{\text{lat}}(\sigma) = \mathcal{E}_{\text{const}} - \frac{\Gamma_{\text{eff}}^2}{4\beta_{\text{lat}}}. \quad (17)$$

Accordingly, the rigorous bounds evaluated at these surrogate bandwidths retain the remainder terms:

$$W_2^2(k_{\sigma_{\text{amb}}^*} * \hat{\mu}_n, \mu) \leq W_0^2 - \frac{\alpha_n^2}{4\beta_{\text{amb}}} + \mathcal{R}_{\text{amb}}(\sigma_{\text{amb}}^*), \quad (18)$$

$$W_2^2(\mu_{\text{out}}(\sigma_{\text{lat}}^*), \mu) \leq \mathcal{E}_{\text{const}} - \frac{\Gamma_{\text{eff}}^2}{4\beta_{\text{lat}}} + \mathcal{R}_{\text{lat}}(\sigma_{\text{lat}}^*). \quad (19)$$

Thus, σ_{amb}^* and σ_{lat}^* are minimizers of the quadratic surrogates, not necessarily minimizers of the exact Wasserstein errors.

Because $\Gamma_{\text{eff}} > 0$ whenever $\rho_{\parallel} > 0$, the negative linear term in Ψ_{lat} dominates its quadratic term for sufficiently small positive σ . More precisely,

$$\Psi_{\text{lat}}(\sigma) < \Psi_{\text{lat}}(0) = \mathcal{E}_{\text{const}} \quad \text{whenever } 0 < \sigma < \frac{\Gamma_{\text{eff}}}{\beta_{\text{lat}}}.$$

The same conclusion holds for the complete asymptotic upper bound for all sufficiently small σ , since $\mathcal{R}_{\text{lat}}(\sigma) = \mathcal{O}(\sigma^3)$. This establishes an initial decrease of the derived upper bound relative to its constant term. It does not, by itself, imply that the exact Wasserstein error is strictly decreasing from its exact value at $\sigma = 0$.

At the level of the second-order surrogates, latent smoothing gives the smaller optimized value,

$$\min_{\sigma \geq 0} \Psi_{\text{lat}}(\sigma) < \min_{\sigma \geq 0} \Psi_{\text{amb}}(\sigma),$$

provided that

$$\mathcal{E}_{\text{const}} - W_0^2 < \frac{\Gamma_{\text{eff}}^2}{4\beta_{\text{lat}}} - \frac{\alpha_n^2}{4\beta_{\text{amb}}}. \quad (20)$$

This is a comparison between the optimized quadratic surrogates. A strict comparison between the complete upper bounds additionally requires the gap in (20) to dominate the remainder terms in (18) and (19).

The excess constant in the latent surrogate is

$$\mathcal{E}_{\text{const}} - W_0^2 = \varepsilon_{\text{iso}} W_0^2 + 2W_0\delta + \delta^2.$$

In particular, $\varepsilon_{\text{iso}} W_0^2$ is introduced by the Young inequality used in the proof and is not the exact zero-noise transport error of the autoencoder.

In the high-dimensional regime, where $\beta_{\text{amb}} \asymp D$ and $D \gg m$, the ambient surrogate improvement $\alpha_n^2/(4\beta_{\text{amb}})$ becomes small. Neglecting this term, the surrogate comparison (20) reduces to

$$\varepsilon_{\text{iso}} W_0^2 + 2W_0\delta + \delta^2 < \frac{\rho_{\parallel}^2 C_m^2 n^{-2/m} \left(1 + \frac{\delta}{W_0}\right)^2}{4 \left(L_{\text{orth}}^2 (d - m) + \varepsilon_{\text{iso}}^2 m + Q\right)}. \quad (21)$$

Condition (21) is a limiting surrogate condition obtained by dropping the ambient improvement term. For finite D , it is necessary but not sufficient for (20). More precisely, define the margin

$$\Delta := \frac{\Gamma_{\text{eff}}^2}{4\beta_{\text{lat}}} - (\mathcal{E}_{\text{const}} - W_0^2).$$

The finite-dimensional surrogate comparison holds exactly when

$$\Delta > \frac{\alpha_n^2}{4\beta_{\text{amb}}}.$$

Thus, a positive margin independent of D guarantees the surrogate comparison for sufficiently large D when $\beta_{\text{amb}} \asymp D$ and the remaining quantities are fixed.

For example, in the exact-reconstruction case $\delta = 0$, condition (21) is equivalent to the following implicit condition on the isometry defect, with Q evaluated at $\delta = 0$:

$$\varepsilon_{\text{iso}} < \frac{\rho_{\parallel}^2 C_m^2 n^{-2/m}}{4W_0^2 \left(L_{\text{orth}}^2 (d - m) + \varepsilon_{\text{iso}}^2 m + Q\right)}. \quad (22)$$

This condition is implicit because ε_{iso} also appears in the second-order coefficient and may enter Q . If, in addition, the empirical error satisfies

$$W_0 \asymp C_{\text{emp}} n^{-1/m}$$

in the dimension and sampling regime under consideration, then $n^{-2/m}/W_0^2$ is of constant order. This cancels the explicit sample-size dependence in that ratio. An admissible isometry defect that remains bounded away from zero as n increases additionally requires control of the other coefficients. For fixed m , d , μ , and $\rho_{\parallel} > 0$, with L_{orth} and the decoder derivative bounds uniformly bounded in n , the condition permits a sufficiently small positive isometry defect independent of n .

Similarly, if the isometry contribution and the quadratic term δ^2 are controlled separately, and if the dependence of β_{lat} on δ is negligible in the regime $\delta \ll W_0$, isolating the interaction term $2W_0\delta$ gives the scaling requirement

$$\delta \lesssim \frac{\rho_{\parallel}^2 C_m^2 n^{-2/m}}{8W_0 (L_{\text{orth}}^2 (d-m) + \varepsilon_{\text{iso}}^2 m + Q)}. \quad (23)$$

Under the additional empirical scaling $W_0 \asymp C_{\text{emp}} n^{-1/m}$, this becomes

$$\delta = \mathcal{O} \left(\frac{\rho_{\parallel}^2 n^{-1/m}}{L_{\text{orth}}^2 (d-m) + \varepsilon_{\text{iso}}^2 m + Q} \right).$$

Thus, the second-order surrogate analysis indicates that favorable latent smoothing requires both controlled tangential distortion and reconstruction error, with increasingly restrictive behavior when the latent-complement penalty $L_{\text{orth}}^2 (d-m)$ is large. These relations describe the parameter dependence of the quadratic surrogate and should not be interpreted as exact formulas for the optimizer of the full Wasserstein error.

4 Proof of Main Theorems

This section proves the three main theoretical results. The proof of Theorem 1 separates the ambient smoothing error into an off-manifold projection term, an intrinsic–extrinsic smoothing discrepancy, and the intrinsic estimation error. The proof of Theorem 2 studies the evolution of the intrinsic Wasserstein error along the heat flow. Finally, the proof of Theorem 3 quantifies the effect of smoothing in a lower-dimensional latent space.

Throughout this section, k_{σ}^M denotes the intrinsic heat kernel at diffusion time $t = \sigma^2$, normalized so that its tangent-space covariance at time t is $t\mathbf{I}_m$. Equivalently, the heat generator is $\frac{1}{2}\Delta_M$. When the diffusion time itself is used as the subscript, k_t^M is shorthand for $k_{\sqrt{t}}^M$. Thus $k_t^M * \hat{\mu}_n$ and $k_{\sigma}^M * \hat{\mu}_n$ refer to the same heat flow under the relation $t = \sigma^2$.

4.1 Proof of Theorem 1: Ambient Smoothing Error Bound

Rather than comparing $k_{\sigma} * \hat{\mu}_n$ directly with μ , we isolate three geometric contributions

1. the displacement caused by projecting ambient Gaussian noise to tangent spaces of M ;
2. the discrepancy between extrinsically projected noise and intrinsic heat smoothing;
and
3. the intrinsic estimation error on M .

The lemmas below control these terms in the order in which they enter the final coupling argument.

To establish a baseline bound on the intrinsic error before introducing ambient noise leakage or latent distortion, we analyze the distance between the ambiently smoothed measure and its metric projection onto tangent space of each point.

Lemma 1 (Extrinsic Projection Distance Bound). *Assume $\tau > \sigma\sqrt{D}$. Let $Y \sim k_\sigma * \hat{\mu}_n$ be the ambiently smoothed empirical distribution on $M \subset \mathbb{R}^D$ with reach $\tau > 0$, and let $\mu_\sigma^{\text{tan}} = \frac{1}{n} \sum_i (\pi_{x_i + T_{x_i} M}) \# \mathcal{N}(x_i, \sigma^2 \mathbf{I}_D)$ denote the tangentially smoothed distribution. The squared 2-Wasserstein distance is bounded by*

$$W_2^2(k_\sigma * \hat{\mu}_n, \mu_\sigma^{\text{tan}}) \leq \sigma^2(D - m).$$

Proof Under the suboptimal coupling induced by the projection map $Y \mapsto \pi_{X_n + T_{X_n} M}(Y)$ with $X_n \sim \hat{\mu}_n$, we bound W_2^2 by the expected squared displacement in ambient space

$$W_2^2(k_\sigma * \hat{\mu}_n, \mu_\sigma^{\text{tan}}) \leq \mathbb{E} [\|Y - \pi_{X_n + T_{X_n} M}(Y)\|^2].$$

Expressing $Y = X_n + W$ and ambient Gaussian noise $W \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_D)$, we decompose $W = W^\parallel + W^\perp$ into orthogonal projections onto the tangent space $T_{X_n} M \cong \mathbb{R}^m$ and normal space $N_{X_n} M \cong \mathbb{R}^{D-m}$. Then,

$$\mathbb{E} [\|Y - \pi_{X_n + T_{X_n} M}(Y)\|^2] = \mathbb{E} [\|W^\perp\|^2] = \sigma^2(D - m).$$

This completes the proof. ■

To complete the intrinsic error channel, we analyze the local behavior of the true intrinsic heat-smoothed distribution $\mu_\sigma = k_\sigma^M * \hat{\mu}_n$ and bridge the gap to its tangential Gaussian approximation μ_σ^{tan} . The following lemma constructs a synchronous coupling between Riemannian Brownian motion on M and its flat tangent-space approximation, establishing explicit L^2 bounds on both the total diffusion displacement $\|X - Z_n\|_{L^2}$ from the initial sample Z_n and the approximation discrepancy $\|X - \tilde{X}\|_{L^2}$ as functions of the intrinsic dimension m , reach τ , and scale parameter σ .

Lemma 2 (Intrinsic Heat Flow vs. Tangent Gaussian Coupling). *Let $M \subset \mathbb{R}^D$ be a compact m -dimensional smooth submanifold with reach $\tau > 0$, and let $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ be an empirical measure on M . Let $X \sim \mu_\sigma = k_\sigma^M * \hat{\mu}_n$ denote the endpoint at diffusion time $t = \sigma^2$ of Riemannian Brownian motion initialized at $Z_n \sim \hat{\mu}_n$. Let $\tilde{X} = Z_n + W^\parallel$ with $W^\parallel \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{T_{Z_n} M})$. Under the synchronous coupling driven by ambient Brownian motion, if $0 < \sigma \leq \frac{\tau}{\sqrt{m}}$, then*

$$\|X - Z_n\|_{L^2} \leq \left(\sqrt{m} + \frac{2m\sigma}{\tau} \right) \sigma, \quad \|X - \tilde{X}\|_{L^2} \leq \frac{2m}{\tau} \sigma^2. \quad (24)$$

Proof Condition on $Z_n = x \in M$. Let $P(y) : \mathbb{R}^D \rightarrow T_y M$ denote the orthogonal projection onto the tangent space at y . The extrinsic representation of Riemannian Brownian motion with generator $\frac{1}{2} \Delta_M$ is

$$X_t = x + \int_0^t P(X_s) dW_s + \frac{1}{2} \int_0^t H(X_s) ds,$$

where W_t is standard Brownian motion in $\mathbb{R}^D$ and H is the mean-curvature vector of the embedding. Adding and subtracting $P(x)W_t$ gives

$$X_t = x + P(x)W_t + \int_0^t (P(X_s) - P(x)) dW_s + \frac{1}{2} \int_0^t H(X_s) ds. \quad (25)$$

We first establish a short-time second-moment bound for $X_s - x$. Set

$$u(s) := \mathbb{E}\|X_s - x\|^2.$$

Applying Itô's formula to the function $q \mapsto \|q - x\|^2$ gives

$$u(s) = \int_0^s [m + \mathbb{E}[\langle X_r - x, H(X_r) \rangle]] dr,$$

and thus

$$u'(s) = m + \mathbb{E}[\langle X_s - x, H(X_s) \rangle].$$

A reach of τ implies $\|H(y)\| \leq \frac{m}{\tau}$ for all $y \in M$. It also implies that for $a, b \in M$ and every unit normal $n \in N_a M$,

$$|\langle b - a, n \rangle| \leq \frac{\|b - a\|^2}{2\tau}.$$

Applying this inequality at $a = X_s$ with $n = H(X_s)/\|H(X_s)\|$ when $H(X_s) \neq 0$ yields

$$|\langle X_s - x, H(X_s) \rangle| \leq \frac{m}{2\tau^2} \|X_s - x\|^2.$$

Consequently,

$$u'(s) \leq m + \frac{m}{2\tau^2} u(s), \quad u(0) = 0.$$

Gronwall's inequality gives

$$u(s) \leq 2\tau^2 \left(\exp\left(\frac{ms}{2\tau^2}\right) - 1 \right).$$

For $0 \leq s \leq \tau^2/m$, we have $\exp\left(\frac{ms}{2\tau^2}\right) - 1 \leq \frac{ms}{\tau^2}$, and therefore

$$\mathbb{E}\|X_s - x\|^2 \leq 2ms. \tag{26}$$

The reach bound also gives the tangent-projection estimate in Frobenius norm

$$\|P(y_1) - P(y_2)\|_F^2 \leq \frac{2m}{\tau^2} \|y_1 - y_2\|^2, \quad y_1, y_2 \in M.$$

Hence, by Itô's isometry and (26),

$$\begin{aligned} \left\| \int_0^t (P(X_s) - P(x)) dW_s \right\|_{L^2}^2 &= \int_0^t \mathbb{E} [\|P(X_s) - P(x)\|_F^2] ds \\ &\leq \frac{2m}{\tau^2} \int_0^t \mathbb{E}\|X_s - x\|^2 ds \\ &\leq \frac{2m}{\tau^2} \int_0^t 2ms ds = \frac{2m^2}{\tau^2} t^2. \end{aligned} \tag{27}$$

For the mean-curvature term, Minkowski's integral inequality gives

$$\left\| \frac{1}{2} \int_0^t H(X_s) ds \right\|_{L^2} \leq \frac{1}{2} \int_0^t \|H(X_s)\|_{L^2} ds \leq \frac{m}{2\tau} t. \tag{28}$$

Combining (27) and (28) in (25), we have

$$\|X_t - x\|_{L^2} \leq \|P(x)W_t\|_{L^2} + \left(\sqrt{2} + \frac{1}{2}\right) \frac{m}{\tau} t \leq \sqrt{mt} + \frac{2m}{\tau} t$$

where the last inequality uses $P(x)W_t \sim \mathcal{N}(0, t\mathbf{I}_{T_x M})$ and $\sqrt{2} + \frac{1}{2} \leq 2$.

Define $\tilde{X}_t := x + P(x)W_t$. Then,

$$X_t - \tilde{X}_t = \int_0^t (P(X_s) - P(x)) dW_s + \frac{1}{2} \int_0^t H(X_s) ds. \quad (29)$$

Combining (27) and (28) in (29), we obtain

$$\|X_t - \tilde{X}_t\|_{L^2} \leq \left(\sqrt{2} + \frac{1}{2}\right) \frac{m}{\tau} t \leq \frac{2m}{\tau} t.$$

The condition $\sigma \leq \tau/\sqrt{m}$ ensures that $t = \sigma^2 \leq \tau^2/m$, so the preceding short-time estimates apply. Setting $t = \sigma^2$ and then averaging over $Z_n \sim \hat{\mu}_n$ concludes the proof. $\blacksquare$

We are now ready to prove Theorem 1. For completeness, we first restate Theorem 1 with explicit constants and non-asymptotic residual terms before giving the proof.

Theorem 1 (Extrinsic W_2^2 Error Decomposition under Ambient Smoothing). *Let $M \subset \mathbb{R}^D$ be an m -dimensional compact submanifold with reach $\tau > 0$. Let μ be a smooth probability measure supported on M , and let $k_\sigma * \hat{\mu}_n$ denote the ambient Gaussian-smoothed empirical measure with bandwidth parameter $\sigma > 0$ satisfying $0 < \sigma < \tau/\sqrt{mD}$. Let $\mu_\sigma = k_\sigma^M * \hat{\mu}_n$ denote the intrinsic heat-smoothed measure on M .*

Then, the squared 2-Wasserstein distance satisfies

$$\begin{aligned} W_2^2(k_\sigma * \hat{\mu}_n, \mu) &\leq W_2^2(\mu_\sigma, \mu) + \sigma^2(D - m) + \frac{4m}{\tau} W_2(\hat{\mu}_n, \mu) \sigma^2 \\ &\quad + \frac{4m}{\tau} (\sqrt{D - m} + \sqrt{m}) \sigma^3 + \frac{12m^2}{\tau^2} \sigma^4 + C_{\text{tail}}, \end{aligned}$$

where C_{tail} is the explicit nonnegative tail bound defined in (31).

Proof Let $Z_n \sim \hat{\mu}_n$. Conditionally on $Z_n = x \in M$, let $(B_s)_{s \geq 0}$ be a standard Brownian motion in $\mathbb{R}^D$ and set $W^\parallel := P(x)B_{\sigma^2} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{T_x M})$, where $P(x)$ denotes the orthogonal projection onto $T_x M$. Independently of the entire Brownian path $(B_s)_{s \geq 0}$, sample $W^\perp \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{N_x M})$. Define $\tilde{X} := x + W^\parallel$ and $Z := x + W^\parallel + W^\perp$. Because $T_x M$ and $N_x M$ are orthogonal complementary subspaces and $W^\parallel$ and $W^\perp$ are independent, conditionally on $Z_n = x$, $W^\parallel + W^\perp \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_D)$. Consequently, $Z \sim k_\sigma * \hat{\mu}_n$. Moreover, $\tilde{X}$ is the orthogonal projection of Z onto the affine tangent plane $x + T_x M$:

$$\tilde{X} = \pi_{x+T_x M}(Z) = x + P(x)(Z - x).$$

Thus $(Z, \tilde{X})$ is the ambient-to-tangent coupling used in Lemma 1.

Using the same Brownian path $(B_s)_{s \geq 0}$ that defines $W^\parallel = P(x)B_{\sigma^2}$, construct the Riemannian Brownian motion $(X_s)_{s \geq 0}$ initialized at $X_0 = x$ as in Lemma 2, and set $X := X_{\sigma^2}$. Then $X \sim \mu_\sigma$, and Lemma 2 gives

$$\|X - \tilde{X}\|_{L^2} \leq \frac{2m}{\tau} \sigma^2.$$

By construction, conditionally on $Z_n = x$, the normal Gaussian vector $W^\perp$ is independent of $(\tilde{X}, X)$.

Finally, let $\pi^\star$ be an optimal coupling of μ_σ and μ for the ambient Euclidean cost. Disintegrate it as

$$\pi^\star(dq, dy) = \mu_\sigma(dq) K(q, dy).$$

After constructing X , sample Y conditionally according to $K(X, dy)$, using randomness independent of $W^\perp$. Then $Y \sim \mu$ and

$$\mathbb{E}\|X - Y\|^2 = W_2^2(\mu_\sigma, \mu).$$

Furthermore, conditionally on $Z_n = x$, $W^\perp$ is independent of $(\tilde{X}, X, Y)$ and satisfies

$$\mathbb{E}[W^\perp \mid Z_n = x] = 0.$$

Next, we decompose the total expectation over the event $\mathcal{E} = \{\|W\| < \tau/\sqrt{m}\}$ and its complement

$$\mathbb{E}[\|Z - Y\|^2] = \mathbb{E}[\|Z - Y\|^2 \mathbf{1}_\mathcal{E}] + \mathbb{E}[\|Z - Y\|^2 \mathbf{1}_{\mathcal{E}^c}]. \quad (30)$$

Since $\mathcal{E}^c = \{\|W\| \geq \tau/\sqrt{m}\}$, the inequality $\|Z - Y\|^2 \leq 2\|W\|^2 + 2\text{diam}(M)^2$ and Proposition 3 give

$$\begin{aligned} \mathbb{E}[\|Z - Y\|^2 \mathbf{1}_{\mathcal{E}^c}] &\leq 2\mathbb{E}[\|W\|^2 \mathbf{1}(\|W\| \geq \tau/\sqrt{m})] + 2\text{diam}(M)^2 \mathbb{P}(\|W\| \geq \tau/\sqrt{m}) \\ &\leq 2\sqrt{3}\sigma^2 D \exp\left(-\frac{(\tau/\sqrt{m} - \sigma\sqrt{D})^2}{4\sigma^2}\right) \\ &\quad + 2\text{diam}(M)^2 \exp\left(-\frac{(\tau/\sqrt{m} - \sigma\sqrt{D})^2}{2\sigma^2}\right) \\ &= C_{\text{tail}}(\tau, \sigma, m, D, M). \end{aligned} \quad (31)$$

On $\mathcal{E}$, we expand the squared Euclidean distance into three direct terms and three interaction terms

$$\begin{aligned} \|Z - Y\|^2 \mathbf{1}_\mathcal{E} &= \left(\|Z - \tilde{X}\|^2 + \|\tilde{X} - X\|^2 + \|X - Y\|^2 \right) \mathbf{1}_\mathcal{E} \\ &\quad + \underbrace{2\langle Z - \tilde{X}, \tilde{X} - X \rangle \mathbf{1}_\mathcal{E}}_{T_1 \mathbf{1}_\mathcal{E}} + \underbrace{2\langle \tilde{X} - X, X - Y \rangle \mathbf{1}_\mathcal{E}}_{T_2 \mathbf{1}_\mathcal{E}} + \underbrace{2\langle Z - \tilde{X}, X - Y \rangle \mathbf{1}_\mathcal{E}}_{T_3 \mathbf{1}_\mathcal{E}}. \end{aligned} \quad (32)$$

Taking expectations over $\mathcal{E}$, the three squared distance terms are bounded as follows

$$\begin{aligned} \mathbb{E}[\|Z - \tilde{X}\|^2 \mathbf{1}_\mathcal{E}] &\leq \sigma^2(D - m) \quad (\text{Lemma 1}), \\ \mathbb{E}[\|\tilde{X} - X\|^2 \mathbf{1}_\mathcal{E}] &\leq \frac{4m^2}{\tau^2} \sigma^4 \quad (\text{Lemma 2}), \\ \mathbb{E}[\|X - Y\|^2 \mathbf{1}_\mathcal{E}] &\leq W_2^2(\mu_\sigma, \mu). \end{aligned}$$

Using Cauchy–Schwarz and the L^2 bound from Lemma 2 gives

$$|\mathbb{E}[T_1 \mathbf{1}_{\mathcal{E}}]| \leq 2\|Z - \tilde{X}\|_{L^2} \|\tilde{X} - X\|_{L^2} \leq \frac{4m\sqrt{D-m}}{\tau} \sigma^3. \quad (33)$$

For the second cross-term T_2 , applying Cauchy–Schwarz gives

$$|\mathbb{E}[T_2 \mathbf{1}_{\mathcal{E}}]| \leq 2\sqrt{\mathbb{E}[\|\tilde{X} - X\|^2]} \sqrt{\mathbb{E}[\|X - Y\|^2]} \leq \frac{4m}{\tau} \sigma^2 W_2(\mu_{\sigma}, \mu). \quad (34)$$

By the triangle inequality and Lemma 2,

$$W_2(\mu_{\sigma}, \mu) \leq W_2(\hat{\mu}_n, \mu) + \sqrt{m} \sigma + \frac{2m}{\tau} \sigma^2.$$

Substituting into (34) yields

$$|\mathbb{E}[T_2 \mathbf{1}_{\mathcal{E}}]| \leq \frac{4m}{\tau} W_2(\hat{\mu}_n, \mu) \sigma^2 + \frac{4m\sqrt{m}}{\tau} \sigma^3 + \frac{8m^2}{\tau^2} \sigma^4.$$

For the third cross-term, condition on $(Z_n, W^{\parallel}, X, Y)$. The normal noise $W^{\perp}$ remains a centered Gaussian under this conditioning. Writing $W = W^{\parallel} + W^{\perp}$, orthogonality gives $\mathcal{E} = \{\|W^{\parallel}\|^2 + \|W^{\perp}\|^2 < \tau^2/m\}$. Gaussian symmetry yields

$$\mathbb{E}[W^{\perp} \mathbf{1}_{\mathcal{E}} \mid Z_n, W^{\parallel}, X, Y] = 0,$$

and therefore $\mathbb{E}[T_3 \mathbf{1}_{\mathcal{E}}] = 0$. Summing the bounds for the direct terms, cross terms, and tail contribution completes the proof. $\blacksquare$

The bound retains the realized empirical error $W_2(\hat{\mu}_n, \mu)$ explicitly and does not require an empirical convergence-rate estimate. A bound in expectation or with high probability can be obtained by combining this result with an empirical Wasserstein estimate valid under the relevant dimensional and distributional assumptions.

4.2 Proof of Theorem 2: Derivative of Intrinsic Wasserstein Distance under Heat Flow

To quantify the statistical benefit of intrinsic smoothing, we study $W_{2,M}^2(\mu_t, \mu)$ along the Riemannian heat flow $\partial_t \mu_t = \frac{1}{2} \Delta_M \mu_t$. As $t \rightarrow 0^+$, or equivalently $\sigma \rightarrow 0^+$ with $t = \sigma^2$, the heat-smoothed measure converges weakly to $\hat{\mu}_n$.

The proof has three steps. We first differentiate the optimal transport cost in the time variable t . We then bound the curvature and Jacobian terms using the transport expansion and the Monge–Ampère equation. Finally, we integrate in time and substitute $t = \sigma^2$. The differentiation step uses the regularity of the optimal maps and Kantorovich potentials supplied by the stated nonfocal and Ma–Trudinger–Wang hypotheses (Ma et al., 2005; Figalli, 2010).

For completeness, we restate Theorem 2 with explicit geometric constants prior to presenting its formal proof.

Theorem 2 (Intrinsic Wasserstein Distance Derivative and Integrated Bounds). *Let (M, g) be a compact, connected, m -dimensional Riemannian manifold without boundary. Assume that M has a nonfocal cut locus and that the squared geodesic cost $c(x, y) = \frac{1}{2}d_M^2(x, y)$ satisfies the MTW(K) condition for some $K > 0$. Let $\mu_t = k_t^M * \hat{\mu}_n$ denote its intrinsic heat-kernel smoothing at diffusion time $t > 0$, where $\partial_t \mu_t = \frac{1}{2}\Delta_M \mu_t$. For the bandwidth parameter $\sigma > 0$, write $\mu_t := \mu_{\sqrt{t}}$ for simplicity of notation.*

Assume that, for every sufficiently small $t > 0$, the optimal transport from μ_t to μ satisfies the regularity conclusions required in Proposition 1. Then

$$\frac{d}{dt} W_{2,M}^2(\mu_t, \mu) \leq -\frac{1}{2} C_m n^{-1/m} t^{-1/2} + m, \quad (35)$$

where

$$C_m := \frac{m \rho_{\max}^{-1/m}}{\sqrt{2\pi}} \left(\frac{m}{m+1} \right)^{m/2}. \quad (36)$$

In particular,

$$\lim_{t \rightarrow 0^+} \frac{d}{dt} W_{2,M}^2(\mu_t, \mu) = -\infty.$$

Integrating (35) and setting $t = \sigma^2$, we obtain, for all sufficiently small $\sigma > 0$,

$$W_{2,M}^2(\mu_\sigma, \mu) \leq W_{2,M}^2(\hat{\mu}_n, \mu) - C_m n^{-1/m} \sigma + m \sigma^2. \quad (37)$$

Before proving the theorem, we establish a pointwise Laplacian bound for the dual Kantorovich potential. The argument combines the differential factorization of Riemannian optimal transport maps (Cordero-Erausquin et al., 2001), the second-variation formula, and Rauch comparison. The strict MTW assumption supplies the nonnegative sectional curvature used by the comparison argument.

Proposition 1 (Laplacian Bound for the Dual Kantorovich Potential). *Assume the same manifold and measure assumptions as in Theorem 2. Let ψ_t be a smooth dual Kantorovich potential associated with the optimal transport from $\mu_t \text{dvol}_M$ to μdvol_M , and let $T_t(y) = \exp_y(-\nabla_M \psi_t(y))$ be the corresponding smooth optimal transport map. Assume that $T_t(y) \notin \text{Cut}(y)$. Then*

$$\Delta_M \psi_t(y) \leq m - m \left(\frac{\mu_t(y)}{\rho_{\max}} \right)^{1/m}. \quad (38)$$

Proof Set $z = T_t(y)$, $v = -\nabla_M \psi_t(y)$, so that $z = \exp_y(v)$. Since $z \notin \text{Cut}(y)$, the function $\frac{1}{2}d_M^2(\cdot, z)$ is smooth near y . Throughout this proof, we identify the Riemannian Hessian of a smooth function h with the self-adjoint endomorphism of $T_y M$ characterized by

$$g_y((\text{Hess}_y h)u, w) = (\nabla^2 h)_y(u, w), \quad u, w \in T_y M.$$

Thus traces, determinants, and compositions below are taken for endomorphisms of $T_y M$. Define

$$H_y := \text{Hess}_y \left(\frac{1}{2} d_M^2(\cdot, z) \right) \quad \text{and} \quad B_t(y) := H_y - \text{Hess}_y \psi_t.$$

With our dual convention, $\phi_t(x) + \psi_t(q) \leq c(x, q)$, with equality at $(x, q) = (z, y)$; thus ψ_t is c -concave. Holding z fixed, the function $G_z(q) := c(z, q) - \phi_t(z) - \psi_t(q)$ has a local minimum at y . Consequently,

$$g_y(B_t(y)w, w) = (\nabla^2 G_z)_y(w, w) \geq 0, \quad w \in T_y M,$$

so $B_t(y)$ is self-adjoint and positive semidefinite.

For the squared Riemannian distance cost, the diagonal value of the MTW cross-curvature recovers the Riemannian sectional curvature, up to a positive normalization factor. Hence the assumed MTW(K) condition with $K > 0$ implies, in particular, nonnegative sectional curvature (Loeper, 2009).

We first show that

$$H_y \leq \mathbf{I}_{T_y M} \tag{39}$$

as quadratic forms. Let

$$\gamma(s) = \exp_y(sv), \quad 0 \leq s \leq 1,$$

be the minimizing geodesic from y to z . For $w \in T_y M$, let J be the Jacobi field along γ satisfying

$$J(0) = w, \quad J(1) = 0.$$

By the second-variation formula for the energy,

$$\langle H_y w, w \rangle = I_\gamma(J, J),$$

where

$$I_\gamma(V, V) := \int_0^1 (\|D_s V\|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle) ds$$

is the index form along γ (Jost, 2005, Section 6.1).

Let $P_{0 \rightarrow s} : T_y M \rightarrow T_{\gamma(s)} M$ denote parallel transport along γ , and consider the dummy vector field

$$V(s) := (1 - s)P_{0 \rightarrow s}w.$$

It has the same endpoint values as J : $V(0) = w$ and $V(1) = 0$. Because γ contains no conjugate point before z , the Jacobi field J minimizes the index form among vector fields with these endpoint values. Therefore,

$$I_\gamma(J, J) \leq I_\gamma(V, V).$$

Since $P_{0 \rightarrow s}w$ is parallel, $D_s V = -P_{0 \rightarrow s}w$, $\|D_s V\| = \|w\|$. Moreover, the nonnegative sectional curvature consequence of the MTW($K > 0$) assumption implies

$$\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle \geq 0.$$

It follows that

$$\langle H_y w, w \rangle = I_\gamma(J, J) \leq I_\gamma(V, V) \leq \int_0^1 \|D_s V\|^2 ds = \|w\|^2.$$

This proves (39), and hence

$$\mathrm{Tr}(H_y) \leq m. \quad (40)$$

We next obtain a lower bound for $\det B_t(y)$. Let $P_v : T_z M \rightarrow T_y M$ denote parallel transport from z back to y , and define

$$Y_y := P_v \circ d(\exp_y)_v \in \mathrm{End}(T_y M).$$

The differential factorization of the optimal transport map (Cordero-Erausquin et al., 2001) gives

$$\tilde{d}T_t(y) := P_v \circ d(T_t)_y = Y_y B_t(y). \quad (41)$$

The same nonnegative sectional curvature consequence implies, by Jacobi Field comparison (Lee, 2018, Theorem 11.9), that

$$\|Y_y w\| \leq \|w\|, \quad w \in T_y M.$$

Since $z \notin \mathrm{Cut}(y)$, the operator Y_y is nonsingular. More explicitly, for $s \in [0, 1]$, define

$$Y_y(s) := P_{s \rightarrow 0} \circ d(\exp_y)_{sv},$$

where $P_{s \rightarrow 0}$ is parallel transport along the geodesic segment $\gamma(s) = \exp_y(sv)$ back to y . Then $Y_y(0) = I_{T_y M}$ and $Y_y(1) = Y_y$. Because the minimizing segment contains no conjugate points, $Y_y(s)$ remains non-singular for all $s \in [0, 1]$. By continuity, $\det Y_y(s)$ cannot change sign, which implies $\det Y_y > 0$. Since $\|Y_y w\| \leq \|w\|$, all singular values of Y_y are at most one. Therefore

$$0 < \det Y_y \leq 1. \quad (42)$$

The Monge–Ampère equation for T_t gives

$$\mu_t(y) = \mu(T_t(y)) \mathrm{Jac}(T_t)(y). \quad (43)$$

Because parallel transport is an isometry,

$$\mathrm{Jac}(T_t)(y) = \sqrt{\det(dT_t(y)^* dT_t(y))} = \sqrt{\det(\tilde{d}T_t(y)^* \tilde{d}T_t(y))} = |\det(\tilde{d}T_t(y))|.$$

The strict positivity of μ_t and μ , together with (43), gives $\mathrm{Jac}(T_t)(y) > 0$. On the other hand, Equation (41), $\det(Y_y) > 0$, and the positive semidefiniteness of $B_t(y)$ show that $\det(\tilde{d}T_t(y)) \geq 0$. Since its absolute value equals the strictly positive Jacobian, it follows that $\det(\tilde{d}T_t(y)) > 0$. Consequently,

$$\mathrm{Jac}(T_t)(y) = \det(\tilde{d}T_t(y)). \quad (44)$$

Using Equations (41) to (44), we obtain

$$\det B_t(y) = \frac{\det(\tilde{d}T_t(y))}{\det(Y_y)} \geq \det(\tilde{d}T_t(y)) = \frac{\mu_t(y)}{\mu(T_t(y))} \geq \frac{\mu_t(y)}{\rho_{\max}}.$$

Again because $B_t(y)$ is positive semidefinite, the arithmetic-geometric mean inequality applied to its eigenvalues gives

$$\mathrm{Tr}(B_t(y)) \geq m (\det B_t(y))^{1/m} \geq m \left(\frac{\mu_t(y)}{\rho_{\max}} \right)^{1/m}.$$

Finally, from the definition of B_t , we have

$$\Delta_M \psi_t(y) = \mathrm{Tr}(H_y) - \mathrm{Tr}(B_t(y)). \quad (45)$$

Combining this identity with Equations (40) and (45) gives

$$\Delta_M \psi_t(y) \leq m - m \left(\frac{\mu_t(y)}{\rho_{\max}} \right)^{1/m},$$

which proves (38). ■

Now that we have established the Laplacian bound for the dual Kantorovich potential, we can proceed to prove Theorem 2.

Proof [Proof of Theorem 2] Set

$$F(t) := W_{2,M}^2(\mu_t, \mu).$$

For the cost $c(x, y) = \frac{1}{2} d_M^2(x, y)$, the Kantorovich dual formulation gives

$$\frac{1}{2} F(t) = \int_M \phi_t(x) \mu(x) \, \mathrm{dvol}(x) + \int_M \psi_t(y) \mu_t(y) \, \mathrm{dvol}(y),$$

where ϕ_t and ψ_t are dual Kantorovich potentials.

Under the nonfocal cut-locus and MTW($K > 0$) assumptions, together with the stated density bounds, the dual potentials and the optimal transport map have the regularity required for the following differentiation (Figalli, 2010). Applying the envelope theorem and using

$$\partial_t \mu_t = \frac{1}{2} \Delta_M \mu_t,$$

we obtain

$$\frac{1}{2} F'(t) = \int_M \psi_t(y) \partial_t \mu_t(y) \, \mathrm{dvol}(y) = \frac{1}{2} \int_M \psi_t(y) \Delta_M \mu_t(y) \, \mathrm{dvol}(y).$$

Since M is compact and has no boundary, integration by parts gives

$$\int_M \psi_t \Delta_M \mu_t \, \mathrm{dvol} = \int_M \mu_t \Delta_M \psi_t \, \mathrm{dvol}.$$

Therefore,

$$F'(t) = \int_M \Delta_M \psi_t(y) \mu_t(y) \, \mathrm{dvol}(y). \quad (46)$$

By Proposition 1, for μ_t -almost every $y \in M$,

$$\Delta_M \psi_t(y) \leq m - m \left(\frac{\mu_t(y)}{\rho_{\max}} \right)^{1/m}.$$

Substituting this estimate into (46) yields

$$F'(t) \leq m - m \rho_{\max}^{-1/m} \int_M \mu_t(y)^{1+1/m} d\text{vol}(y). \quad (47)$$

Since

$$\mu_t(y) = \frac{1}{n} \sum_{i=1}^n k_t^M(y, x_i)$$

and the function $s \mapsto s^{1+1/m}$ is superadditive on $[0, \infty)$, we have

$$\mu_t(y)^{1+1/m} \geq \frac{1}{n^{1+1/m}} \sum_{i=1}^n k_t^M(y, x_i)^{1+1/m}.$$

Integrating over M gives

$$\int_M \mu_t(y)^{1+1/m} d\text{vol}(y) \geq \frac{1}{n^{1+1/m}} \sum_{i=1}^n \int_M k_t^M(y, x_i)^{1+1/m} d\text{vol}(y). \quad (48)$$

Choose a fixed radius $r_0 > 0$ smaller than the injectivity radius of M . In normal coordinates $y = \exp_x(v)$ with $\|v\| < r_0$, the small-time heat-kernel expansion gives

$$k_t^M(\exp_x(v), x) = (2\pi t)^{-m/2} \exp\left(-\frac{\|v\|^2}{2t}\right) (u_0(x, v) + \mathcal{O}(t)),$$

where $u_0(x, v) = 1 + \mathcal{O}(\|v\|^2)$. The Riemannian volume element in these coordinates is

$$d\text{vol}_M(\exp_x(v)) = (1 + \mathcal{O}(\|v\|^2)) dv.$$

The estimates are uniform in x by compactness. Restricting the integral to this normal neighborhood and rescaling $v = \sqrt{t} z$, the spatial correction contributes a relative $\mathcal{O}(t)$ term. The omitted Euclidean Gaussian tail is exponentially small. Consequently, uniformly in $x \in M$,

$$\int_M k_t^M(y, x)^{1+1/m} d\text{vol}(y) \geq \frac{1}{\sqrt{2\pi}} \left(\frac{m}{m+1} \right)^{m/2} t^{-1/2} (1 - C_M t)$$

for sufficiently small $t > 0$. Using this in (48), we obtain

$$\int_M \mu_t(y)^{1+1/m} d\text{vol}(y) \geq \frac{n^{-1/m}}{\sqrt{2\pi}} \left(\frac{m}{m+1} \right)^{m/2} t^{-1/2} (1 - C_M t). \quad (49)$$

Decrease $t_0 > 0$, if necessary, so that

$$1 - C_M t \geq \frac{1}{2} \quad \text{for every } 0 < t < t_0.$$

It follows that

$$\int_M \mu_t(y)^{1+1/m} \mathrm{dvol}(y) \geq \frac{n^{-1/m}}{2\sqrt{2\pi}} \left(\frac{m}{m+1} \right)^{m/2} t^{-1/2}. \quad (50)$$

Substituting (50) into (47), and using the definition (36), gives

$$F'(t) \leq -\frac{1}{2} C_m n^{-1/m} t^{-1/2} + m.$$

This proves (35). Since the right-hand side converges to $-\infty$ as $t \rightarrow 0^+$, we also obtain

$$\lim_{t \rightarrow 0^+} F'(t) = -\infty.$$

The heat semigroup converges weakly to its initial measure, and its second moments remain uniformly bounded because M is compact. Therefore,

$$\mu_t \longrightarrow \hat{\mu}_n \quad \text{in } W_{2,M}$$

as $t \rightarrow 0^+$. Consequently,

$$F(t) \longrightarrow W_{2,M}^2(\hat{\mu}_n, \mu).$$

Integrating (35) from 0 to t gives

$$\begin{aligned} F(t) &\leq W_{2,M}^2(\hat{\mu}_n, \mu) - \frac{1}{2} C_m n^{-1/m} \int_0^t s^{-1/2} ds + m \int_0^t ds \\ &= W_{2,M}^2(\hat{\mu}_n, \mu) - C_m n^{-1/m} \sqrt{t} + mt. \end{aligned}$$

Setting $t = \sigma^2$ yields

$$W_{2,M}^2(\mu_\sigma, \mu) \leq W_{2,M}^2(\hat{\mu}_n, \mu) - C_m n^{-1/m} \sigma + m\sigma^2.$$

This completes the proof. ■

Remark 2 (The round sphere). *While the global nonfocal cut-locus assumption in Theorem 2 provides a sufficient condition to guarantee the regularity of optimal transport maps, it is not strictly necessary in every geometry. The round sphere $M = \mathbb{S}_R^m$ fails this condition because its cut locus consists of conjugate antipodal points. Nevertheless, for the squared geodesic distance cost on the sphere, the strong MTW condition holds outside the antipodal cut locus, and optimal maps between smooth, strictly positive densities are known to stay away from this cut locus, remaining globally C^∞ smooth (Loeper, 2011). In our setting, μ_t and μ are smooth and strictly positive for all $t > 0$, ensuring that the first-variation argument applies on every subinterval $[\varepsilon, t_0]$ with $\varepsilon > 0$. Integrating over $[\varepsilon, t_0]$ and taking the limit $\varepsilon \rightarrow 0^+$ (using $\mu_\varepsilon \rightarrow \hat{\mu}_n$ in $W_{2,M}$) yields the exact integrated estimate of Theorem 2. Consequently, the round-sphere experiment in Section 5 falls within a regime where the required transport regularity is fully established via sphere-specific regularity theory.*

4.3 Proof of Theorem 3: Dimension Reduction via Latent-Space Smoothing

The combined ambient bound contains the dimensional term $D\sigma^2 = m\sigma^2 + (D - m)\sigma^2$, together with geometric corrections. The decomposition distinguishes the intrinsic contribution from the additional normal-noise contribution.

Latent smoothing introduces noise in $\mathbb{R}^d$ and maps it back through the decoder. Its explicit orthogonal contribution is $L_{\text{orth}}^2(d - m)\sigma^2$, while tangential gain, isometry defect, reconstruction error, and geometry also enter the quadratic coefficient. Thus the comparison uses the full coefficients β_{lat} and β_{amb} , rather than codimensions alone. The decoder bounds quantify when reducing the noise dimension can yield a tighter optimized surrogate.

For completeness, we restate Theorem 3 with all explicit geometric conditions, local Jacobian bounds, and non-asymptotic residual terms before presenting the proof.

Theorem 3 (Latent-space smoothing bound). *Let M be a compact, connected, m dimensional Riemannian manifold embedded in $\mathbb{R}^D$ with reach $\tau > 0$. Assume the hypotheses of Theorem 2, or the round-sphere setting of Remark 2, so that the intrinsic smoothing bound applies.*

Let $f : \mathbb{R}^D \rightarrow \mathbb{R}^d$ and $g : \mathbb{R}^d \rightarrow \mathbb{R}^D$, where $m \leq d \leq D$, and define δ as in Equation (7). Assume that f is differentiable at each x_i and that $g \in C^2(\mathbb{R}^d; \mathbb{R}^D)$ has uniformly bounded first and second derivatives. Set

$$C_g := \max \left\{ 1, \sup_{z \in \mathbb{R}^d} \|J_g(z)\|_{\text{op}}, \sup_{z \in \mathbb{R}^d} \|D^2 g(z)\|_{\text{op}} \right\} < \infty. \quad (51)$$

For each i , assume that

$$U_i := J_f(x_i)|_{T_{x_i}M} : T_{x_i}M \rightarrow V_i$$

is a linear isometry onto an m -dimensional subspace $V_i \subset \mathbb{R}^d$, and define

$$A_i := J_g(f(x_i))J_f(x_i)|_{T_{x_i}M} : T_{x_i}M \rightarrow \mathbb{R}^D.$$

Assume that there exist a global tangential gain $\rho_{\parallel} > 0$ and an effective isometry defect $\varepsilon_{\text{iso}} \geq 0$ such that

$$\varepsilon_{\text{iso}} = \max_{1 \leq i \leq n} \min_{O \in O(T_{x_i}M)} \|A_i - \rho_{\parallel} \iota_i O\|_{\text{op}}, \quad (52)$$

where $\iota_i : T_{x_i}M \hookrightarrow \mathbb{R}^D$ is the inclusion map. Let O_i be a minimizing orthogonal map for each i . Assume also that

$$\|J_g(f(x_i))v^{\perp}\| \leq L_{\text{orth}}\|v^{\perp}\|, \quad v^{\perp} \in V_i^{\perp}.$$

Define $\mu_{\text{out}} := g_{\#}((f_{\#}\hat{\mu}_n) * \mathcal{N}(0, \sigma^2 \mathbf{I}_d))$ and $W_0 := W_{2,M}(\hat{\mu}_n, \mu)$. For sufficiently small $\sigma > 0$ such that $\rho_{\parallel}\sigma$ lies in the small-bandwidth regime of Theorem 2 and $0 < \sigma < \frac{1}{2\sqrt{m}\rho_{\parallel}} \min\{W_0, 2\tau\}$, one has

$$W_2^2(\mu_{\text{out}}, \mu) \leq \mathcal{E}_{\text{const}} - \Gamma_{\text{eff}}\sigma + L_{\text{latent}}\sigma^2 + \mathcal{O}(\sigma^3), \quad (53)$$

where

$$\begin{aligned}\mathcal{E}_{\text{const}} &:= (1 + \varepsilon_{\text{iso}})W_0^2 + 2\delta W_0 + \delta^2, \\ \Gamma_{\text{eff}} &:= \rho_{\parallel} C_m n^{-1/m} \left(1 + \frac{\delta}{W_0}\right), \\ L_{\text{latent}} &:= L_{\text{orth}}^2(d - m) + \varepsilon_{\text{iso}}^2 m + Q.\end{aligned}$$

Here C_m is the constant from Theorem 2, and Q is defined in Equation (54). The constant implicit in $\mathcal{O}(\sigma^3)$ may depend on the fixed geometric, empirical, and functions regularities appearing above, but is independent of σ .

Proof Define the following quantities:

$$\begin{aligned}C_{\text{geom}} &:= \frac{2m}{\tau}, \quad G_2 := \frac{C_g}{2} \sqrt{d(d+2)}, \quad \gamma_n := C_m n^{-1/m}, \\ Q &:= \rho_{\parallel}^2 m \left(1 + \varepsilon_{\text{iso}} + \frac{\delta}{W_0}\right) + m\varepsilon_{\text{iso}} + 2 \left(C_{\text{geom}} \rho_{\parallel}^2 + G_2\right) (W_0 + \delta).\end{aligned}\tag{54}$$

Let

$$M_i := A_i - \rho_{\parallel} \iota_i O_i, \quad \|M_i\|_{\text{op}} \leq \varepsilon_{\text{iso}},$$

and set

$$r := \rho_{\parallel} \sigma, \quad \mu_r := k_r^M * \hat{\mu}_n, \quad W_r := W_{2,M}(\mu_r, \mu).$$

We define a joint coupling to estimate $W_2^2(\mu_{\text{out}}, \mu)$. Let $X_n \sim \hat{\mu}_n$, so that $X_n = x_i$ with probability $1/n$. Conditionally on $X_n = x_i$, sample $\xi \sim \mathcal{N}(0, \mathbf{I}_d)$ and decompose it orthogonally as

$$\xi = \xi^{\parallel} + \xi^{\perp}, \quad \xi^{\parallel} \in V_i, \quad \xi^{\perp} \in V_i^{\perp}.$$

Since $U_i : T_{x_i} M \rightarrow V_i$ is an isometry,

$$W^{\parallel} := U_i^{-1} \xi^{\parallel} \sim \mathcal{N}(0, \mathbf{I}_{T_{x_i} M}).$$

The rotated vector $v := O_i W^{\parallel}$ has the same distribution.

By Lemma 2, we may couple the flat tangent step $x_i + r \iota_i v$ to a point z_i with conditional law $k_r^M(x_i, \cdot)$ so that

$$\|z_i - (x_i + r \iota_i v)\|_{L^2} \leq C_{\text{geom}} r^2.\tag{55}$$

After averaging over i , the variable z_i has law μ_r . Disintegrate an optimal coupling for the intrinsic cost between μ_r and μ , and conditionally on z_i sample y from this coupling. Then

$$\mathbb{E}[d_M^2(z_i, y)] = W_r^2$$

and, since the ambient Euclidean distance is bounded by the intrinsic distance on M ,

$$\|y - z_i\|_{L^2} \leq W_r.\tag{56}$$

Conditionally on $X_n = x_i$, the orthogonal noise $\xi^{\perp}$ is independent of (z_i, y) .

Let

$$Z_{\text{out}} := g(f(x_i) + \sigma \xi).$$

We use the same four-term decomposition as in the original argument:

$$\begin{aligned}
 y - Z_{\text{out}} &= (y - z_i) + (z_i - (x_i + r\iota_i v)) \\
 &\quad + ((x_i + r\iota_i v) - (g(f(x_i)) + r\iota_i v)) \\
 &\quad + ((g(f(x_i)) + r\iota_i v) - g(f(x_i) + \sigma\xi)) \\
 &=: T_1 + T_2 + T_3 + T_4.
 \end{aligned} \tag{57}$$

Writing

$$d_i := g(f(x_i)) - x_i,$$

the third term is simply

$$T_3 = -d_i, \quad \mathbb{E}\|T_3\|^2 = \delta^2.$$

Taylor's theorem and Equation (51) give

$$g(f(x_i) + \sigma\xi) = g(f(x_i)) + \sigma J_g(f(x_i))\xi + R_g(\sigma\xi),$$

where

$$\|R_g(\sigma\xi)\| \leq \frac{C_g}{2} \sigma^2 \|\xi\|^2. \tag{58}$$

Since $\xi^\parallel = U_i W^\parallel$,

$$J_g(f(x_i))\xi^\parallel = A_i W^\parallel = \rho_\parallel \iota_i v + M_i W^\parallel.$$

Consequently,

$$T_4 = -\sigma M_i W^\parallel - \sigma J_g(f(x_i))\xi^\perp - R_g(\sigma\xi). \tag{59}$$

We first estimate the four squared terms. From Equations (55) and (56),

$$\mathbb{E}\|T_1\|^2 \leq W_r^2, \quad \mathbb{E}\|T_2\|^2 \leq C_{\text{geom}}^2 r^4, \quad \mathbb{E}\|T_3\|^2 = \delta^2.$$

The two linear terms in Equation (59) are orthogonal in expectation because $W^\parallel$ and $\xi^\perp$ are independent and centered conditionally on $X_n = x_i$. Moreover,

$$\mathbb{E}\|M_i W^\parallel\|^2 \leq \varepsilon_{\text{iso}}^2 m, \quad \mathbb{E}\|J_g(f(x_i))\xi^\perp\|^2 \leq L_{\text{orth}}^2 (d - m).$$

The Gaussian identity $\mathbb{E}\|\xi\|^4 = d(d+2)$ and Equation (58) imply

$$\|R_g(\sigma\xi)\|_{L^2} \leq G_2 \sigma^2.$$

It follows that

$$\mathbb{E}\|T_4\|^2 \leq (\varepsilon_{\text{iso}}^2 m + L_{\text{orth}}^2 (d - m)) \sigma^2 + \mathcal{O}(\sigma^3). \tag{60}$$

We next estimate the six cross terms. By Cauchy-Schwarz,

$$2\mathbb{E}\langle T_1, T_2 \rangle \leq 2C_{\text{geom}} r^2 W_r, \quad 2\mathbb{E}\langle T_1, T_3 \rangle \leq 2\delta W_r.$$

The component of $2\mathbb{E}\langle T_1, T_4 \rangle$ containing $\xi^\perp$ vanishes by conditional independence and centering. For the tangential defect, Cauchy-Schwarz and Young's inequality give

$$2\mathbb{E}\langle y - z_i, -\sigma M_i W^\parallel \rangle \leq 2\varepsilon_{\text{iso}} \sqrt{m} \sigma W_r \leq \varepsilon_{\text{iso}} W_r^2 + m\varepsilon_{\text{iso}} \sigma^2. \tag{61}$$

The Taylor remainder satisfies

$$2\mathbb{E}\langle y - z_i, -R_g(\sigma\xi) \rangle \leq 2G_2\sigma^2 W_r.$$

The remaining cross terms satisfy

$$\begin{aligned} 2\mathbb{E}\langle T_2, T_3 \rangle &\leq 2\delta C_{\text{geom}} r^2, \\ 2\mathbb{E}\langle T_2, T_4 \rangle &= \mathcal{O}(\sigma^3), \\ 2\mathbb{E}\langle T_3, T_4 \rangle &\leq 2\delta G_2 \sigma^2. \end{aligned}$$

For the last line, the two linear components of T_4 have zero conditional mean given $X_n = x_i$, so only the Taylor remainder contributes.

Combining the direct and cross-term estimates yields

$$\begin{aligned} W_2^2(\mu_{\text{out}}, \mu) &\leq (1 + \varepsilon_{\text{iso}})W_r^2 + 2\delta W_r + \delta^2 \\ &\quad + \left[\varepsilon_{\text{iso}}^2 m + m\varepsilon_{\text{iso}} + L_{\text{orth}}^2(d - m) \right] \sigma^2 \\ &\quad + 2 \left(C_{\text{geom}} \rho_{\parallel}^2 + G_2 \right) (W_r + \delta) \sigma^2 + \mathcal{O}(\sigma^3). \end{aligned} \tag{62}$$

We now use the intrinsic estimate. By Theorem 2,

$$W_r^2 \leq W_0^2 - \gamma_n r + m r^2. \tag{63}$$

Because $W_0 > 0$, concavity of the square-root function gives

$$W_r \leq W_0 - \frac{\gamma_n}{2W_0} r + \frac{m}{2W_0} r^2. \tag{64}$$

Finally, in the term already multiplied by σ^2 in Equation (62), we may replace W_r by W_0 ; the resulting difference is $\mathcal{O}(\sigma^3)$. Substituting $r = \rho_{\parallel} \sigma$ into Equation (62) and using the preceding bounds initially gives the favorable linear coefficient $-\rho_{\parallel} \gamma_n (1 + \varepsilon_{\text{iso}} + \delta/W_0)$. Discarding the additional nonpositive term $-\rho_{\parallel} \gamma_n \varepsilon_{\text{iso}} \sigma$ yields the slightly weaker but simpler bound

$$\begin{aligned} W_2^2(\mu_{\text{out}}, \mu) &\leq (1 + \varepsilon_{\text{iso}})W_0^2 + 2\delta W_0 + \delta^2 - \rho_{\parallel} \gamma_n \left(1 + \frac{\delta}{W_0} \right) \sigma \\ &\quad + \left[L_{\text{orth}}^2(d - m) + \varepsilon_{\text{iso}}^2 m + \rho_{\parallel}^2 m \left(1 + \varepsilon_{\text{iso}} + \frac{\delta}{W_0} \right) \right. \\ &\quad \left. + m\varepsilon_{\text{iso}} + 2 \left(C_{\text{geom}} \rho_{\parallel}^2 + G_2 \right) (W_0 + \delta) \right] \sigma^2 + \mathcal{O}(\sigma^3). \end{aligned}$$

Recalling $\gamma_n = C_m n^{-1/m}$ and the definitions of $\mathcal{E}_{\text{const}}$, Γ_{eff} , L_{latent} , and Q proves Equation (53).

Because the first and second derivatives of g are uniformly bounded on all of $\mathbb{R}^d$, the Taylor estimate Equation (58) holds for every Gaussian realization. Thus no truncation event or separate Gaussian tail term is needed under the assumptions of the theorem. $\blacksquare$

5 Numerical Experiments

We conduct numerical experiments on synthetic manifolds to evaluate the theoretical predictions of Theorems 1 to 3. Specifically, we examine the interaction among intrinsic variance reduction, ambient-dimensional inflation, and autoencoder reconstruction fidelity. Throughout the experiments, we use the sphere as a model geometry for the theoretical results, as discussed in Remark 2. We also consider a manifold outside the stated geometric regime to investigate whether the predicted behavior may persist beyond the assumptions of the theorems. The scripts for reproducing all experimental results are available at https://github.com/wonjune/intrinsic_smoothing.

5.1 Ambient Smoothing and Intrinsic Variance Reduction

In this section, we will numerically validate the dependence of the ambient smoothing error $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ and the optimal bandwidth σ_{amb}^* on the sample size n and ambient dimension D , using synthetic manifold data and discrete optimal transport computation. In particular, the quadratic surrogate Ψ_{amb} (12) results in the following scaling

$$\sigma_{\text{amb}}^* = \frac{\alpha_n}{2\beta_{\text{amb}}} \asymp \frac{n^{-1/m}}{D} \iff \log(\sigma_{\text{amb}}^*) = (-1)\log(D) + \left(-\frac{1}{m}\right)\log(n) + \mathcal{O}(1).$$

Section 5.1.1 and 5.1.2 examine the impact of D and n , respectively.

The experiments focus on two MTW($K > 0$) manifolds $\mathbb{S}^3$ and $\mathbb{S}^4$, and one non-MTW($K > 0$) manifold $\mathbb{S}^1 \times \mathbb{S}^2$. The non-MTW($K > 0$) example tests the generality of our predicted scaling. In all these examples, the corresponding target measure μ is taken to be the uniform distribution supported on the manifold. For each manifold tested, we compute the empirical error $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ over 81 quadratically spaced bandwidths $\sigma \in [0, 0.6]$ with

$$\sigma_p = 0.6 \left(\frac{p}{80}\right)^2, \quad p = 0, 1, \dots, 80$$

over five trials for each parameter pair (n, D) .

Optimal transport computation via discretization of marginal distributions

We approximate $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ using $N_{\text{eval}} = 5000$ independent samples from each marginal. Writing $A = k_\sigma * \hat{\mu}_n$, let $\hat{A}_{N_{\text{eval}}}$ and $\hat{\mu}_{N_{\text{eval}}}$ denote the corresponding empirical measures. The reported $\widehat{W}_2^2$ is their discrete Kantorovich cost, computed using the Python POT solver `ot.emd2` (Flamary et al., 2021). To relate this empirical comparison to the underlying distributions, the triangle inequality gives

$$\left| W_2(\hat{A}_{N_{\text{eval}}}, \hat{\mu}_{N_{\text{eval}}}) - W_2(A, \mu) \right| \leq W_2(\hat{A}_{N_{\text{eval}}}, A) + W_2(\hat{\mu}_{N_{\text{eval}}}, \mu).$$

Thus, the discrepancy in W_2 is controlled by the sampling errors of the two marginals. For scale comparison, $N_{\text{eval}}^{-1/m}$ is approximately 0.058 for $m = 3$ and 0.119 for $m = 4$. These nominal intrinsic scales motivate our evaluation size, although they do not provide a quantitative accuracy guarantee: the actual errors depend on the distributions, and intrinsic-dimensional scaling need not apply to the ambient-smoothed marginal.

The following explains how we sample the marginal distributions for each independent trial.

- To discretize μ , if $M = \mathbb{S}^3$ (respectively, $\mathbb{S}^4$), we first sample N_{eval} iid standard Gaussian vectors in $\mathbb{R}^4$ (respectively, in $\mathbb{R}^5$) with covariance $\mathbf{I}_4$ (respectively, $\mathbf{I}_5$) and then normalize each to unit norm, yielding uniform samples on the sphere M . For $M = \mathbb{S}^1 \times \mathbb{S}^2$, we independently sample

$$\{a_i\}_{i=1}^{N_{\text{eval}}} \stackrel{iid}{\sim} \text{Unif}(\mathbb{S}^1), \quad \{b_i\}_{i=1}^{N_{\text{eval}}} \stackrel{iid}{\sim} \text{Unif}(\mathbb{S}^2)$$

and use $\{(a_i, b_i)\}_{i=1}^{N_{\text{eval}}} \subset \mathbb{R}^5$ as iid samples from $\mu = \text{Unif}(M)$. We then embed these low-dimensional samples into $\mathbb{R}^D$ by zero-padding the remaining coordinates, denoting the resulting samples by $\{z_j\}_{j=1}^{N_{\text{eval}}} \stackrel{iid}{\sim} \mu$. The discrete approximation of μ is then

$$\hat{\mu} = \frac{1}{N_{\text{eval}}} \sum_{j=1}^{N_{\text{eval}}} \delta_{z_j}.$$

- To discretize $k_\sigma * \hat{\mu}_n$ at different bandwidth σ_p , we first draw n samples $\{x_i\}_{i=1}^n \stackrel{iid}{\sim} \mu$, embed them into $\mathbb{R}^D$ by zero-padding the remaining coordinates, and form $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. For each empirical measure $\hat{\mu}_n$, we independently draw

$$\{i(j)\}_{j=1}^{N_{\text{eval}}} \stackrel{iid}{\sim} \text{Unif}\{1, 2, \dots, n\}, \quad \{\eta_j\}_{j=1}^{N_{\text{eval}}} \stackrel{iid}{\sim} \mathcal{N}(0, \mathbf{I}_D)$$

and reuse these draws across all bandwidths $\sigma_p \in [0, 0.6]$ to reduce Monte Carlo variability in the resulting W_2^2 curves. For each bandwidth σ_p , we approximate $k_{\sigma_p} * \hat{\mu}_n$ by

$$\hat{\nu}_{\sigma_p} = \frac{1}{N_{\text{eval}}} \sum_{j=1}^{N_{\text{eval}}} \delta_{y_j^{(p)}}, \quad \text{where } y_j^{(p)} = x_{i(j)} + \sigma_p \eta_j, \quad \forall j = 1, 2, \dots, N_{\text{eval}}.$$

We then numerically approximate $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ by $W_2^2(\hat{\nu}_\sigma, \hat{\mu})$ for every $\sigma \in \{\sigma_0, \sigma_1, \dots, \sigma_{80}\}$, computed using the solver `ot.emd2`.

5.1.1 IMPACT OF AMBIENT DIMENSION (D)

To isolate the impact of ambient dimension D , we fix the sample size at $n = 50$ and vary D . We examine the mean $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ error curves as functions of σ and the scaling of σ_{amb}^* with respect to D .

For each of the five trials, we independently draw an empirical measure $\hat{\mu}_n$ and an N_{eval} -point discretization of μ . Both point clouds are held fixed across all D within each trial and embedded into $\mathbb{R}^D$ by zero-padding. For each D , we separately discretize $k_\sigma * \hat{\mu}_n$, following the procedure described above.

Let $C_{t,D}(\sigma)$ denote the computed $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ errors in trial t at ambient dimension D . We define the empirical optimal bandwidth for the mean W_2^2 curve by

$$\hat{\sigma}_{\text{amb}}(D) \in \arg \min_{\sigma \in \{\sigma_0, \dots, \sigma_{80}\}} \frac{1}{5} \sum_{t=1}^5 C_{t,D}(\sigma)$$

as a numerical proxy for $\sigma_{\text{amb}}^*(D)$.

For each manifold tested, we use two collections of ambient dimensions, L_1 and L_2 , for different purposes: L_1 illustrates how $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ varies with bandwidth σ , as well as how the optimal bandwidth and minimum W_2^2 error vary with D , whereas L_2 is used to estimate the slope of $\log(\sigma_{\text{amb}}^*)$ on $\log(D)$. Specifically,

$$L_1 := \{6, 8, 10, 12, 14\}, \quad \text{and} \quad L_2 := \{10 \cdot 2^k : k = 0, 1, \dots, 7\}.$$

Our theory predicts that, for a fixed n ,

$$\sigma_{\text{amb}}^* \asymp \frac{1}{D} \iff \log(\sigma_{\text{amb}}^*) = (-1) \cdot \log(D) + \mathcal{O}(1).$$

Thus, we expect a slope of approximately -1 when regressing $\log(\sigma_{\text{amb}}^*)$ on $\log(D)$.

First, for $D \in L_1$, we plot the mean curves of $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ versus σ , with ± 1 -standard deviation band among 5 trials. Although the experiments cover $\sigma \in [0, 0.6]$, we restrict the displayed curves to $\sigma \in [0, 0.2]$ or $[0, 0.3]$ for better visualization. Figure 1 (top row) shows the results for the two MTW($K > 0$) examples and the non-MTW($K > 0$) example¹. For comparison, reference dashed lines indicate the mean unsmoothed errors $W_2^2(\hat{\mu}_n, \mu)$ and $W_{2,M}^2(\hat{\mu}_n, \mu)$. The results suggest that for a fixed n , increasing D shifts the empirical optimal bandwidth toward zero while increasing the minimum achievable W_2^2 error. Thus, ambient smoothing becomes progressively less effective at reducing $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ as D increases.

Second, for $D \in L_2$, we plot $\hat{\sigma}_{\text{amb}}$ against D on log-log axes and regress $\log(\hat{\sigma}_{\text{amb}})$ on $\log(D)$. Figure 1 (bottom row) gives fitted slopes of -1.0574 for $\mathbb{S}^3$, -1.0494 for $\mathbb{S}^4$, and -0.9711 for $\mathbb{S}^1 \times \mathbb{S}^2$. Each is within 6% of the predicted slope -1 .

5.1.2 IMPACT OF SAMPLE SIZE (n)

To isolate the effect of sample size n , we fix the ambient dimension at $D = 15$ and vary n . We examine the mean $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ error curves as functions of σ and the scaling of σ_{amb}^* with n .

For each of the five trials and each sample size n , we independently draw a new empirical measure $\hat{\mu}_n$ and an N_{eval} -point discretization of μ . We then discretize $k_\sigma * \hat{\mu}_n$ accordingly.

Let $R_{t,n}(\sigma)$ denote the computed $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ errors at the trial t . We define the corresponding optimal bandwidth for the mean $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ curves by

$$\hat{\sigma}_{\text{amb}}(n) := \arg \min_{\sigma_p \in \{\sigma_0, \sigma_1, \dots, \sigma_{80}\}} \frac{1}{5} \sum_{t=1}^5 R_{t,n}(\sigma)$$

as a numerical proxy for $\sigma_{\text{amb}}^*(n)$. Our theory predicts that, for a fixed D ,

$$\sigma_{\text{amb}}^* \asymp n^{-1/m} \iff \log(\sigma_{\text{amb}}^*) = \left(-\frac{1}{m}\right) \cdot \log(n) + \mathcal{O}(1).$$

1. For clarity, we display the mean $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ curves over $\sigma \in [0, 0.20]$ for $\mathbb{S}^3$ and $\mathbb{S}^4$, and over $\sigma \in [0, 0.3]$ for $\mathbb{S}^1 \times \mathbb{S}^2$.

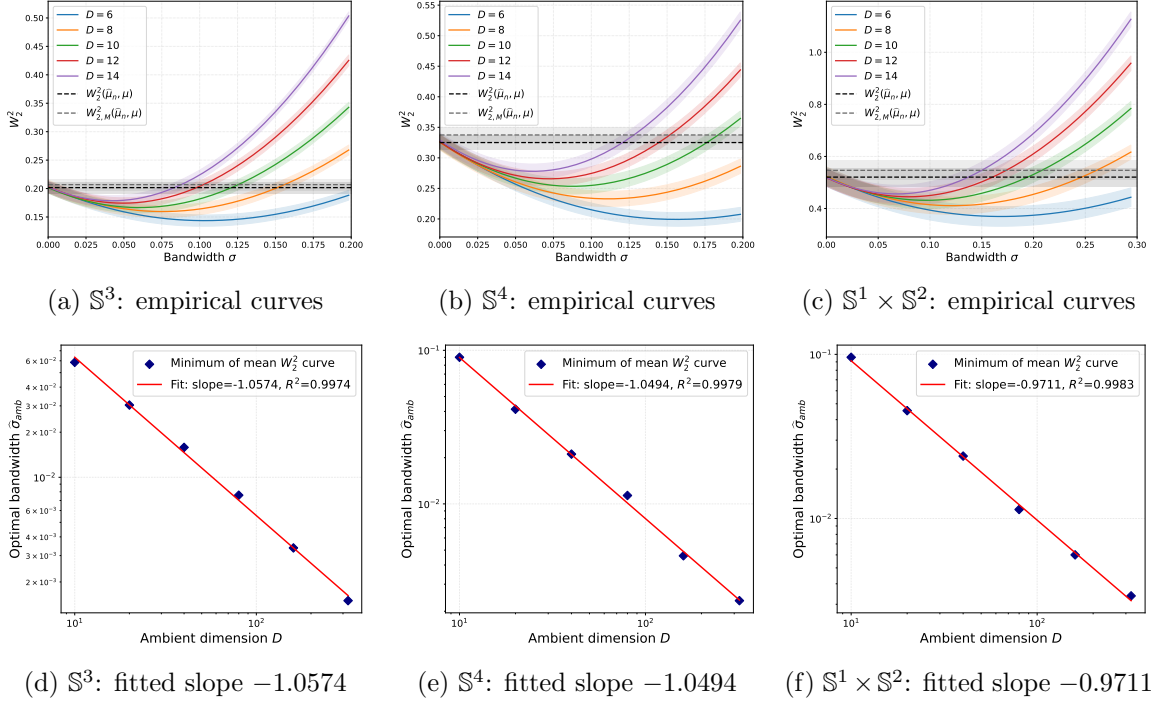


Figure 1: Dependence of ambient smoothing on the ambient dimension D for the MTW($K > 0$) manifolds $\mathbb{S}^3$ and $\mathbb{S}^4$ and the non-MTW($K > 0$) product manifold $\mathbb{S}^1 \times \mathbb{S}^2$. **Top row:** Mean empirical $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ as functions of σ , with ± 1 standard deviation bands over 5 trials. Colors indicate different ambient dimensions, and the horizontal reference lines show the corresponding unsmoothed errors. **Bottom row:** Log-log plots of the empirical bandwidth $\hat{\sigma}_{\text{amb}}$ against $D \in \{10, 20, 40, 80, 160, 320\}$, with linear regression fits in red. From left to right, the fitted slopes are -1.0574 , -1.0494 , and -0.9711 , agreeing with the predicted scaling $\hat{\sigma}_{\text{amb}} \asymp D^{-1}$.

Thus, we expect an approximate slope of $-1/m$ when regressing $\log(\sigma_{\text{amb}}^*)$ on $\log(n)$. To test this prediction, we consider $n \in \{5, 10, 20, 40, 80\}$.

First, we plot the mean $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ against σ , with ± 1 standard deviation bands. Figure 2 (top row) shows the results for all three manifolds². For comparison, the plots include reference dashed lines indicating the mean unsmoothed $W_2^2(\hat{\mu}_n, \mu)$ and $W_{2,M}(\hat{\mu}_n, \mu)$ errors. Colors distinguish different sample sizes n . The results suggest that, increasing n flattens the initial decrease in $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ and shifts the empirical optimal bandwidth toward zero.

Second, we examine the scaling of $\hat{\sigma}_{\text{amb}}$ on n by regressing $\log(\hat{\sigma}_{\text{amb}})$ on $\log(n)$. Figure 2 (bottom row) gives fitted slopes of -0.3361 for $\mathbb{S}^3$, -0.2591 for $\mathbb{S}^4$, and -0.3317 for $\mathbb{S}^1 \times \mathbb{S}^2$. Each is within 4% of its predicted value $-1/m$.

2. For visualization, we display the mean $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ curves over $\sigma \in [0, 0.3]$ in Figure 2 (top row).

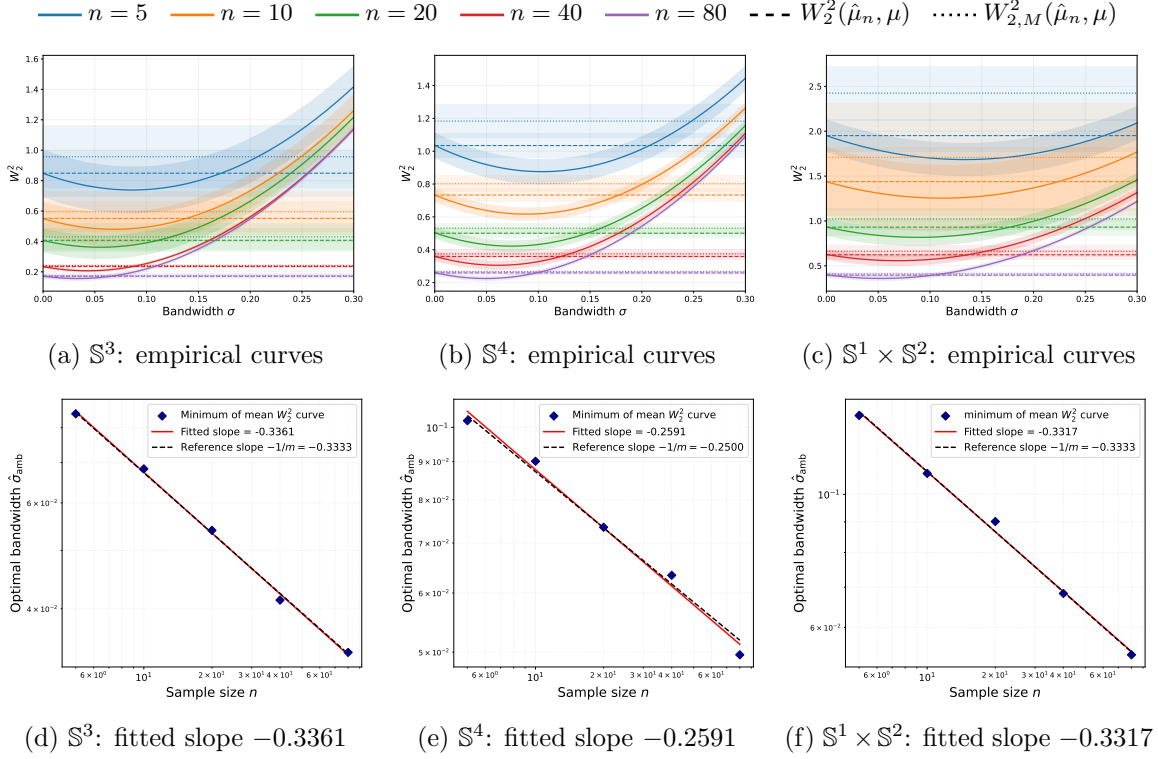


Figure 2: Dependence of ambient smoothing on the empirical sample size n at fixed ambient dimension $D = 15$. Columns correspond, from left to right, to the MTW($K > 0$) manifolds $\mathbb{S}^3$ and $\mathbb{S}^4$ and the non-MTW($K > 0$) product manifold $\mathbb{S}^1 \times \mathbb{S}^2$. **Top row:** Mean empirical $W_2^2(k_\sigma * \hat{\mu}_n, \mu)$ as functions of σ , with ± 1 standard deviation bands over five trials and colors indicating different sample sizes. The dashed horizontal lines indicates the unsmoothed errors $W_2^2(\hat{\mu}_n, \mu)$ and $W_{2,M}^2(\hat{\mu}_n, \mu)$. **Bottom row:** Log-log plots of the empirical bandwidth $\hat{\sigma}_{\text{amb}}$ against n . Red lines show the fitted linear regressions, and the reference lines have the predicted slope $-1/m$. The fitted slopes are -0.3361 , -0.2591 , and -0.3317 , respectively, consistent with $\hat{\sigma}_{\text{amb}} \asymp n^{-1/m}$.

5.2 Latent Smoothing and Encoder Quality Thresholds (Theorem 3)

The experiments in this subsection isolate the four local quantities in Theorem 3: the tangential gain $\rho_{\parallel}$, the orthogonal decoder response L_{orth} , the number $d - m$ of latent directions orthogonal to the encoded tangent space, and the reconstruction error δ . Across both test manifolds, the numerical results show the same basic behavior. The preferred bandwidth increases with $\rho_{\parallel}$ and decreases when either L_{orth} or $d - m$ increases.

These comparisons use explicit global maps f_d and g , rather than inserting the theorem parameters directly into a sampling formula. Their local Jacobians can therefore be calculated analytically and checked by finite differences. The complete Wasserstein curves, including an ambient smoothing reference and a magnified view of the small-bandwidth regime, are reported in Appendix B.

5.2.1 MANIFOLDS AND TARGET DISTRIBUTIONS

We use the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$ and the product $\mathbb{S}^1 \times \mathbb{S}^2 \subset \mathbb{R}^2 \times \mathbb{R}^3$. Both are embedded in the same ambient space $\mathbb{R}^{100}$ by appending zero coordinates. Hence the intrinsic dimensions are $m = 2$ and $m = 3$, respectively, while the ambient dimension is fixed at $D = 100$.

On $\mathbb{S}^2$, we sample from a two-cluster density. For a unit vector $q \in \mathbb{S}^2$ and $\kappa > 0$, let

$$p_{q,\kappa}(x) := \frac{\exp(\kappa \langle q, x \rangle)}{\int_{\mathbb{S}^2} \exp(\kappa \langle q, u \rangle) d\text{vol}(u)}.$$

We take

$$\mu_{\mathbb{S}^2} = 0.62 p_{q_1,8} + 0.38 p_{q_2,11}, \quad q_1 = (0, 0, 1), \quad q_2 = \frac{(0.88, 0.15, -0.45)}{\|(0.88, 0.15, -0.45)\|}.$$

Thus the two components are centered at different points of the sphere and have different concentrations.

For the circle factor, write a point as $(\cos \theta, \sin \theta)$ and set

$$p_{\theta_0,\kappa}(\theta) := \frac{\exp\{\kappa \cos(\theta - \theta_0)\}}{\int_0^{2\pi} \exp\{\kappa \cos(s - \theta_0)\} ds}.$$

We use

$$\mu_{\mathbb{S}^1} = 0.58 p_{0.25,6} + 0.42 p_{3.65,9}, \quad \mu_{\text{prod}} = \mu_{\mathbb{S}^1} \otimes \mu_{\mathbb{S}^2}.$$

The round sphere supplies the positive-curvature example underlying the local mechanism in the theorem. As discussed in Remark 2, the squared-distance cost is used away from the antipodal cut locus. The product $\mathbb{S}^1 \times \mathbb{S}^2$ has flat mixed planes and is included as an out-of-regime stress test, not as an example satisfying the strict MTW($K > 0$) hypothesis.

5.2.2 EXPLICIT ENCODER AND DECODER

For each run, we draw $n = 50$ independent anchors $x_1, \dots, x_n \sim \mu$ and form $\hat{\mu}_n = n^{-1} \sum_{i=1}^n \delta_{x_i}$. Let $d_0 = 3$ on $\mathbb{S}^2$ and $d_0 = 5$ on $\mathbb{S}^1 \times \mathbb{S}^2$. For any $d_0 \leq d \leq D$, the encoder is the coordinate projection

$$f_d(x_1, \dots, x_D) = (x_1, \dots, x_d). \tag{65}$$

Because the final $D - d_0$ coordinates of the embedded manifolds vanish, J_{f_d} preserves every tangent-vector norm exactly.

Let $E_i \in \mathbb{R}^{D \times m}$ contain an orthonormal basis of $T_{x_i}M$, and let $\tilde{E}_i \in \mathbb{R}^{d \times m}$ be obtained by retaining its first d rows. Then $\tilde{E}_i^\top \tilde{E}_i = I_m$. Define

$$c_i = f_d(x_i), \quad P_i = \tilde{E}_i \tilde{E}_i^\top,$$

so P_i projects onto the encoded tangent space. On $\mathbb{S}^2$, the columns of E_i are obtained by normalizing $x_i \times r_i$ and then taking its cross product with x_i . We use $r_i = (1, 0, 0)$ unless this is nearly parallel to x_i , in which case we use $r_i = (0, 1, 0)$. On the product manifold, we add the circle tangent $(-x_{i,2}, x_{i,1})$ to the two sphere tangents.

Let $\iota_d : \mathbb{R}^d \rightarrow \mathbb{R}^D$ be coordinate inclusion and let I_d denote the identity on $\mathbb{R}^d$. The local decoder centered at c_i is

$$h_i(z) := \exp_{x_i}^M \left(\rho_{\parallel} E_i \tilde{E}_i^{\top} (z - c_i) \right) + L_{\text{orth}} \iota_d (I_d - P_i)(z - c_i). \quad (66)$$

For $M = \mathbb{S}^1 \times \mathbb{S}^2$, the exponential map is applied separately to the two factors. To join the local maps, let P_{d_0} denote projection onto the first d_0 coordinates and define the cardinal weights

$$w_i(z) := \frac{\prod_{k \neq i} \|P_{d_0}(z - c_k)\|^4}{\sum_{j=1}^n \prod_{k \neq j} \|P_{d_0}(z - c_k)\|^4}. \quad (67)$$

For distinct anchors, these weights satisfy $w_i(c_i) = 1$, $w_j(c_i) = 0$ for $j \neq i$, and $Dw_j(c_i) = 0$ for every j . Finally, choose a fixed unit vector b and set

$$g(z) := \sum_{i=1}^n w_i(z) h_i(z) + \delta b. \quad (68)$$

In the implementation, b is the unit vector opposite to the population mean. This fixes one reproducible direction for the reconstruction displacement without changing the decoder Jacobian.

Proposition 2 (Prescribed local decoder responses). *The maps in (65) and (68) satisfy, at every empirical anchor,*

$$\|g(f_d(x_i)) - x_i\| = \delta,$$

and

$$J_g(c_i) = \rho_{\parallel} E_i \tilde{E}_i^{\top} + L_{\text{orth}} \iota_d (I_d - P_i).$$

Consequently, for every $v \in T_{x_i} M$ and $v^{\perp} \in \text{range}(P_i)^{\perp}$,

$$J_g(f_d(x_i)) J_{f_d}(x_i) v = \rho_{\parallel} v, \quad \|J_g(f_d(x_i)) v^{\perp}\| = L_{\text{orth}} \|v^{\perp}\|.$$

Thus the construction realizes the tangential gain, reconstruction error, orthogonal response, and latent complement dimension appearing in Theorem 3, with zero tangential isometry defect.

Proof At $z = c_i$, cardinality of the weights gives $g(c_i) = h_i(c_i) + \delta b = x_i + \delta b$. Their first derivatives vanish at the centers, so differentiating only h_i and using $d(\exp_{x_i}^M)_0 = I_{T_{x_i} M}$ gives the displayed Jacobian. Since $\tilde{E}_i^{\top} \tilde{E}_i = I_m$ and P_i is an orthogonal projection, the two response identities follow. $\blacksquare$

5.2.3 NUMERICAL PROTOCOL

For each manifold, we perform five independent runs. Each run draws a new set of $n = 50$ anchors, an independent target sample of size $N_{\text{OT}} = 500$, and new Gaussian coordinates. For a given run, the same anchors, target sample, empirical-mixture indices, and Gaussian coordinates are reused across all values in a parameter sweep. This common-random-number

construction reduces the variability of comparisons within a run without treating repeated evaluations of one empirical measure as independent data.

For every σ , we generate N_{OT} points of the form

$$g(f_d(x_{I_j}) + \sigma \xi_j), \quad I_j \sim \text{Unif}\{1, \dots, n\}, \quad \xi_j \sim \mathcal{N}(0, I_d).$$

The bandwidth grid is

$$\sigma_j = 0.1 \left(\frac{j}{80} \right)^2, \quad j = 0, \dots, 80,$$

which places more grid points near zero. We select the bandwidth grid point with the smallest computed transport cost as the estimated optimal bandwidth.

Optimal-bandwidth plots show means over the five independent runs, with 95% Student- t confidence intervals. Normalized optimal-bandwidth comparisons use ratios of mean optima, with paired first-order ratio intervals. This avoids dividing by an individual run's reference optimum when its estimated minimum occurs at $\sigma = 0$. The decoder identities in Proposition 2 are checked by central finite differences at ten anchors in every run.

5.2.4 TANGENTIAL GAIN

We vary

$$\rho_{\parallel} \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$$

while fixing $\delta = 0$, $L_{\text{orth}} = 1$, and $d - m = 80$. Thus $L_{\text{orth}}^2(d - m) = 80$ is deliberately large relative to the changing tangential terms in the quadratic coefficient. In this regime, the local quadratic minimizer suggested by Theorem 3 has the dominant dependence

$$\frac{\sigma_{\text{quad}}^*(\rho_{\parallel})}{\sigma_{\text{quad}}^*(1)} \approx \rho_{\parallel}. \quad (69)$$

Figure 3 compares the observed normalized minimizers with the slope-one line in (69). The observed points and confidence intervals are plotted first; the theoretical line is drawn last so it remains visible. The figure also reports the slope and confidence interval of an empirical line constrained to pass through (1, 1). For transparency, we include the more general descriptive curve

$$\frac{\sigma_{\text{quad}}^*(\rho_{\parallel})}{\sigma_{\text{quad}}^*(1)} = \frac{\rho_{\parallel}(1 + \beta)}{\rho_{\parallel}^2 + \beta},$$

where the nuisance ratio $\beta \geq 0$ is fitted. This fitted curve is not presented as an independent theoretical prediction.

5.2.5 ORTHOGONAL RESPONSE AND LATENT DIMENSION

We next vary

$$L_{\text{orth}} \in \{0, 0.15, 0.25, 0.35, 0.5\}$$

while fixing $\rho_{\parallel} = 1$, $\delta = 0$, and $d - m = 80$. The theorem isolates the quadratic penalty $L_{\text{orth}}^2(d - m)\sigma^2$. If all remaining second-order terms are summarized by $q_{\text{orth}} \geq 0$, the corresponding local minimizer has the reciprocal form

$$\sigma_{\text{quad}}^*(L_{\text{orth}}) = \frac{a}{80L_{\text{orth}}^2 + q_{\text{orth}}}. \quad (70)$$

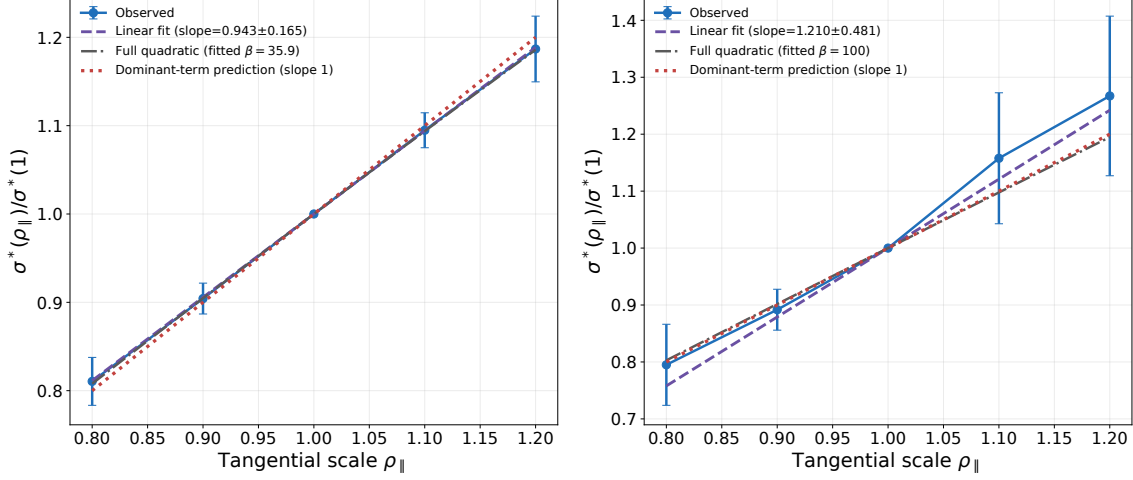


Figure 3: Dependence of the preferred bandwidth on the tangential gain. Left: $\mathbb{S}^2$. Right: $\mathbb{S}^1 \times \mathbb{S}^2$. Points and error bars are the observed mean normalized minimizers and their 95% confidence intervals. The red dotted line is the slope-one dominant-term prediction. The anchored empirical slope is reported in each legend, while the gray curve is a one-parameter descriptive fit.

The gray curve in Figure 4 fits the single nuisance parameter q_{orth} . The red line instead shows the dominant scaling $\sigma^* \propto L_{\text{orth}}^{-2}$, anchored at the largest tested positive value and using no fitted nuisance parameter.

For the dimension experiment, we fix $\rho_{\parallel} = 1$, $\delta = 0$, and $L_{\text{orth}} = 0.5$, and vary

$$d - m \in \{10, 25, 40, 60, 80\}.$$

On $\mathbb{S}^2$, these values correspond to $d \in \{12, 27, 42, 62, 82\}$; on the product, they correspond to $d \in \{13, 28, 43, 63, 83\}$. The explicit penalty is now $0.25(d - m)\sigma^2$. The gray comparison curve has the form

$$\sigma_{\text{quad}}^*(d) = \frac{a}{0.25(d - m) + q_d},$$

with one fitted $q_d \geq 0$, while the red line shows the dominant scaling $\sigma^* \propto (d - m)^{-1}$ anchored at $d - m = 10$. Because the remainder Q in the theorem can also depend on d , this experiment isolates the explicit orthogonal-dimensional contribution rather than claiming a complete formula for all dimension dependence.

The reconstruction-error sweep and its interpretation are reported in Appendix B.1.

Scope of the numerical evidence. The experiment verifies the local parameter identities of the explicit encoder-decoder and tests the parameter dependence of the small-bandwidth bound. It does not establish sharpness of the theorem's constants. The comparison on $\mathbb{S}^1 \times \mathbb{S}^2$ is especially useful in this respect: agreement there suggests that the latent-smoothing mechanism may extend beyond the stated strict MTW regime, but it is not evidence that the current proof applies outside that regime.

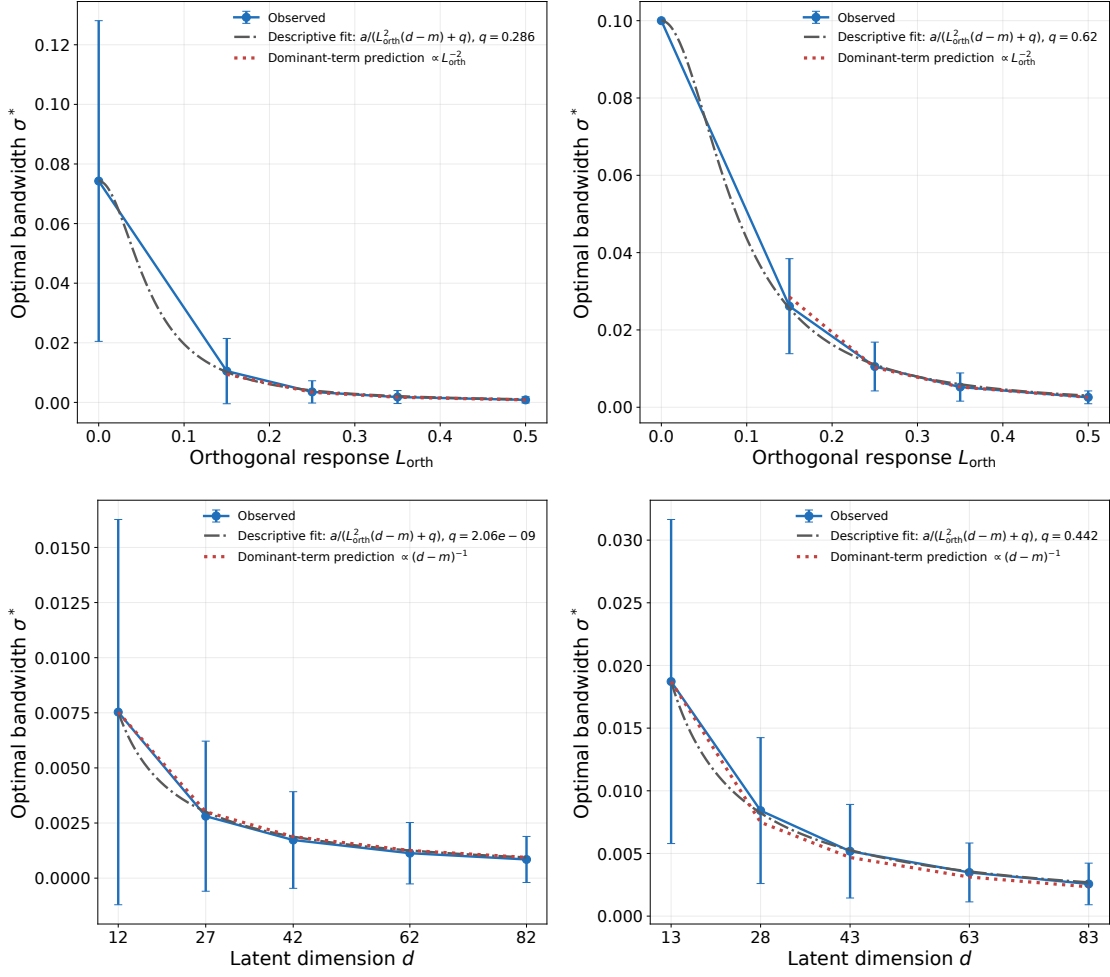


Figure 4: Effect of the orthogonal decoder response and the latent dimension. Top row: L_{orth} . Bottom row: d . Left column: $\mathbb{S}^2$. Right column: $\mathbb{S}^1 \times \mathbb{S}^2$. Observed means and 95% confidence intervals are plotted in blue. Gray curves are one-parameter descriptive reciprocal fits. Red dotted curves are the anchored dominant-term predictions L_{orth}^{-2} and $(d-m)^{-1}$.

5.3 Exploratory Real-Data Study on MNIST

The preceding synthetic experiments evaluate the theoretical quantities in a setting where the manifold and its tangent spaces are known. We now consider MNIST as a complementary real-data example. Each image is represented as a vector in $\mathbb{R}^{784}$, but MNIST does not come with a known smooth manifold, intrinsic dimension, intrinsic metric, or verifiable MTW($K > 0$) structure. Consequently, this experiment is not used to verify the hypotheses of Theorem 3. Instead, it examines whether perturbing a learned low-dimensional representation can provide a more useful smoothing mechanism than perturbing all pixel coordinates directly.

Experimental setup. Using random seed 42, we select a balanced empirical sample of $n = 20$ images, with two images from each digit class. We train the Geometry-Preserving Encoder–Decoder of Lee et al. (2025) using only these anchors, with encoder $f_\theta : \mathbb{R}^{784} \rightarrow \mathbb{R}^{15}$ and decoder $g_\phi : \mathbb{R}^{15} \rightarrow \mathbb{R}^{784}$. The decoder is trained in the presence of latent Gaussian perturbations.

The complete network architecture, training objectives, optimization settings, and local-PCA diagnostics are given in Appendix C. Because MNIST has no known tangent spaces or intrinsic metric, these diagnostics describe the trained network relative to data-estimated PCA frames; they are not interpreted as estimates of the geometric constants in Theorem 3.

Finite-reference transport comparison. For evaluation, let

$$\hat{\mu}_{\text{ref}} = \frac{1}{N_{\text{eval}}} \sum_{j=1}^{N_{\text{eval}}} \delta_{x_j^{\text{ref}}}, \quad N_{\text{eval}} = 5000,$$

be a fixed reference sample from the larger MNIST training collection. The encoder and decoder are not optimized against these reference images. An exact discrete optimal transport calculation between the generated and reference samples in $\mathbb{R}^{784}$ is computationally expensive, so we use the debiased Sinkhorn divergence implemented by GeomLoss (Feydy et al., 2019), with `p=2`, `blur=0.01`, and `scaling=0.8`. The GeomLoss convention for `p=2` uses the ground cost $\frac{1}{2}\|x - y\|_2^2$. We multiply the returned value by two so that its ground-cost normalization agrees with $\|x - y\|_2^2$. We call the resulting entropically regularized quantity the *Sinkhorn transport value*; it is not identified with the exact W_2^2 distance.

Perturbation-scale convention. The numerical scale of a learned latent representation is not fixed a priori. We therefore define

$$s_z := \text{median}_{i < j} \|f_\theta(x_i) - f_\theta(x_j)\|_2$$

and generate latent perturbations according to

$$z_i^\lambda = f_\theta(x_i) + \lambda s_z \zeta, \quad \zeta \sim \mathcal{N}(0, I_{15}).$$

The pixel-space baseline is generated by

$$x_i^\lambda = \text{clamp}(x_i + \lambda \eta, 0, 1), \quad \eta \sim \mathcal{N}(0, I_{784}).$$

Thus λ is a dimensionless experimental control. The corresponding noise standard deviations are λs_z in latent coordinates and λ in pixel coordinates. The common horizontal variable makes it convenient to display both sweeps, but it does not assign the same physical noise scale to the two spaces. The clipped pixel perturbation is also a bounded-image baseline rather than the exact ambient Gaussian convolution appearing in Theorem 1.

5.3.1 SINKHORN TRANSPORT VALUE AND EMPIRICAL COVERAGE

The two perturbation mechanisms are evaluated over the grids shown in Figure 5. For this trained split, the pixel-space baseline reaches a shallow minimum near $\lambda_{\text{pix}} = 0.10$ and then increases rapidly. The latent curve reaches a lower minimum near $\lambda_{\text{lat}} = 0.245$ and grows

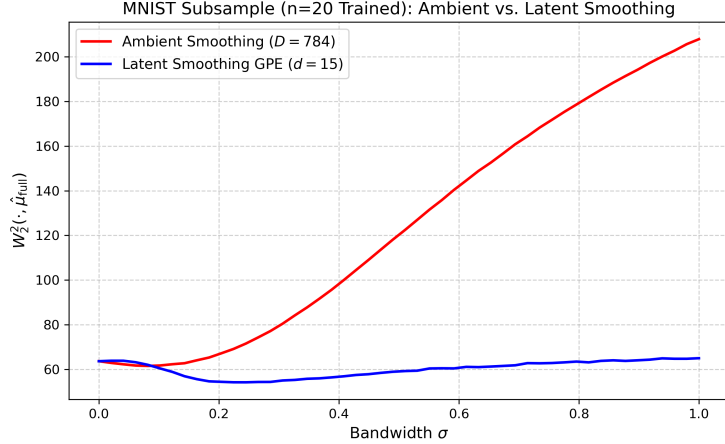


Figure 5: Sinkhorn transport value relative to the finite reference measure $\hat{\mu}_{\text{ref}}$ with $N_{\text{eval}} = 5000$, plotted against the dimensionless perturbation parameter λ . Blue: Gaussian perturbations in the $d = 15$ latent space followed by decoding. Red: Gaussian perturbations in all $D = 784$ pixel coordinates followed by clipping to $[0, 1]^{784}$. The two perturbation mechanisms use the scale conventions specified in the text, so their horizontal coordinates are experimental controls rather than a common physical bandwidth.

more slowly over the displayed range. The corresponding minimum Sinkhorn transport values are approximately 61 and 54, respectively. Because latent and pixel perturbations use different scale normalizations, the locations 0.10 and 0.245 should not be compared as estimates of the same theoretical bandwidth. The relevant observation is that, after optimizing each experimental control separately, the decoded latent samples attain the lower value in this experiment.

The contrast is consistent with the qualitative motivation for latent smoothing. Pixel perturbations act independently in all 784 coordinates. Latent perturbations act in 15 coordinates and are subsequently shaped by the decoder, which was trained to reconstruct the anchors in the presence of latent noise. The experiment does not isolate dimensionality from this learned decoder effect, so the observed improvement should be attributed to the complete latent perturbation and decoding procedure.

To visualize the distributions underlying the transport curves, we project the reference images, training anchors, and generated samples onto a common two-dimensional PCA coordinate system in Figure 6. For each anchor, we generate 500 pixel-space and 500 latent perturbations at their respective empirically selected values of λ . In this projection, the pixel-space samples remain concentrated near the anchors, whereas the decoded latent samples occupy a broader part of the region containing the reference images. This plot describes coverage only after projection and is not used as evidence of tangent or geodesic motion.

The representative images in Figure 7 provide a complementary qualitative comparison. Pixel-space perturbations progressively obscure the digit structure as their magnitude increases. Decoded latent perturbations remain recognizable over a wider displayed range and can produce variations beyond the original anchors.

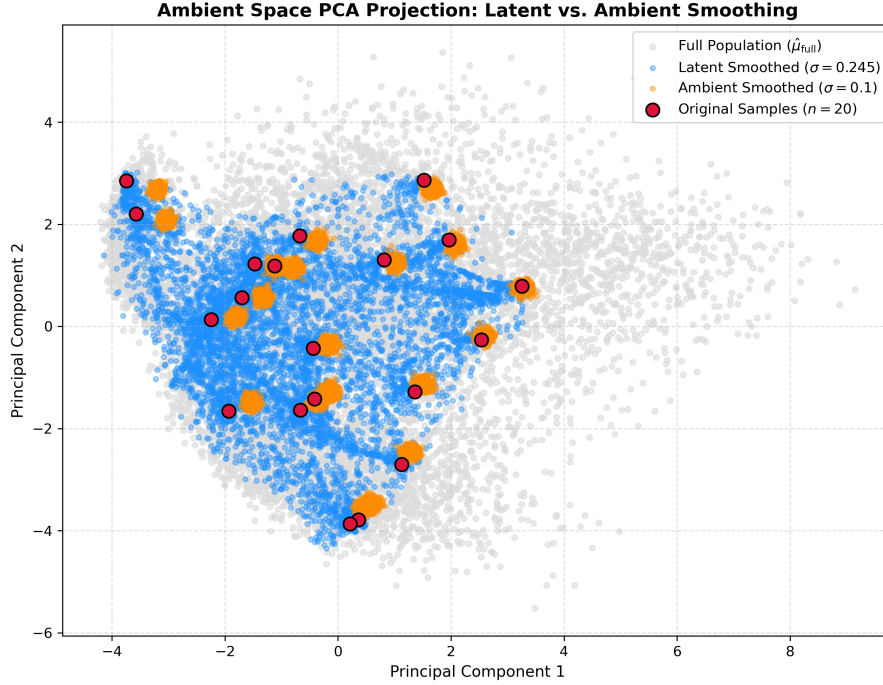


Figure 6: Two-dimensional PCA visualization of the finite reference sample, the $n = 20$ training anchors, and generated samples at the empirically selected perturbation levels. Gray: reference images. Crimson: training anchors. Orange: clipped pixel-space perturbations at $\lambda_{\text{pix}} = 0.10$. Light blue: decoded latent perturbations at $\lambda_{\text{lat}} = 0.245$. The PCA projection is used only to visualize empirical coverage and does not represent intrinsic geodesic coordinates.

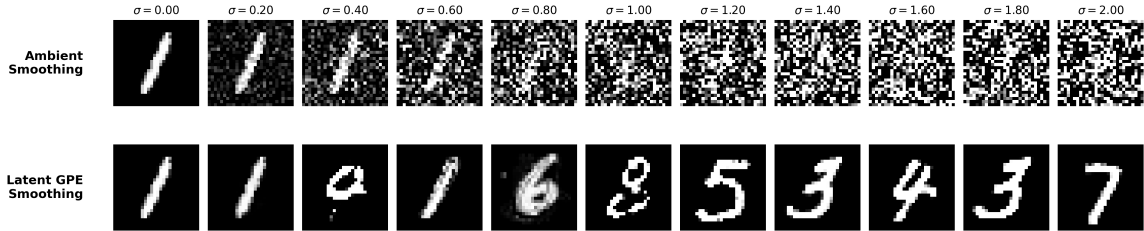


Figure 7: Representative samples at increasing values of the dimensionless perturbation parameter λ . Upper row: Gaussian perturbations in all 784 pixel coordinates followed by clipping. Lower row: Gaussian perturbations in the 15-dimensional latent space followed by decoding. The noise is sampled independently at each displayed value, so columns show representative samples rather than a single coupled trajectory.

Taken together, the Sinkhorn comparison and sample visualizations show that, for this trained model and finite reference sample, perturbing and decoding the 15-dimensional representation performs better than the clipped pixel-space baseline over the tested parameter grids. This real-data result supports the practical motivation for latent smoothing, while

the controlled synthetic experiments remain the setting in which the geometric assumptions and parameter dependence of Theorem 3 are directly tested.

6 Conclusion

In this paper, we studied intrinsic, ambient, and latent smoothing of empirical measures supported on low-dimensional manifolds. Under the stated MTW regularity assumptions, sufficiently small intrinsic heat smoothing improves the squared intrinsic Wasserstein error. Ambient noise also moves mass away from the manifold, and our combined bound explains how the surrounding dimension restricts the smoothing scale it suggests. The latent analysis shows how reconstruction accuracy and the encoder–decoder’s response to perturbations shape this tradeoff.

The resulting second-order surrogates provide local bandwidth-selection rules and sufficient conditions under which the optimized latent surrogate is tighter than its ambient counterpart. These results show that the benefit of latent smoothing depends not only on reducing dimension, but also on how accurately the encoder–decoder pair preserves tangential directions and controls complementary latent directions. The surrogate minimizers should not be interpreted as exact minimizers of the Wasserstein error.

The synthetic experiments support the predicted bandwidth and parameter trends. The product-of-spheres example suggests that some trends may persist beyond the strict MTW regime, although it does not extend the proof. The MNIST experiment is exploratory and shows that decoded latent perturbations achieve a lower Sinkhorn transport value than the pixel-space baseline in the tested setting. Extending the theory to more general manifolds remains an important direction for future work.

Acknowledgments and Disclosure of Funding

Wonjun Lee acknowledges support from a startup fund at The Ohio State University. Computational resources were provided by The Ohio State University.

Appendix A. Appendix

This appendix provides additional technical details, proofs, and numerical diagnostics for the main results.

A.1 Additional Technical Propositions and Missing Proofs

Proposition 3 (Gaussian Tail Integral Bounds). *For any radius $r > \sigma\sqrt{D}$, let $\gamma_\sigma = \mathcal{N}(0, \sigma^2 \mathbf{I}_D)$ denote the isotropic Gaussian measure in $\mathbb{R}^D$ and take $\varepsilon \sim \gamma_\sigma$. The quadratic and zeroth moment tail integrals over the exterior domain $\{\|\varepsilon\| > r\}$ satisfy*

$$\mathbb{E} [\|\varepsilon\|^2 \mathbf{1}(\|\varepsilon\| > r)] \leq \sqrt{3} \sigma^2 D \exp \left(-\frac{(r - \sigma\sqrt{D})^2}{4\sigma^2} \right), \quad (71)$$

$$\mathbb{P}(\|\varepsilon\| > r) \leq \exp \left(-\frac{(r - \sigma\sqrt{D})^2}{2\sigma^2} \right). \quad (72)$$

Proof Let $\varepsilon \sim \gamma_\sigma$. By the Cauchy–Schwarz inequality, $\mathbb{E}\|\varepsilon\| \leq \sqrt{\mathbb{E}\|\varepsilon\|^2} = \sigma\sqrt{D}$. To prove (72),

$$\begin{aligned} \mathbb{P}(\|\varepsilon\| > r) &= \mathbb{P}(\|\varepsilon\| - \sigma\sqrt{D} > r - \sigma\sqrt{D}) \\ &\leq \mathbb{P}(\|\varepsilon\| - \mathbb{E}\|\varepsilon\| > r - \sigma\sqrt{D}) \\ &\leq \exp\left(-\frac{(r - \sigma\sqrt{D})^2}{2\sigma^2}\right), \end{aligned}$$

where the first inequality follows from $\mathbb{E}\|\varepsilon\| \leq \sigma\sqrt{D}$ and the second inequality follows from the Gaussian concentration inequality (Boucheron et al., 2013, Theorem 5.6). To prove (71),

$$\begin{aligned} \mathbb{E}[\|\varepsilon\|^2 \mathbf{1}(\|\varepsilon\| > r)] &= \sum_{i=1}^D \mathbb{E}[\varepsilon_i^2 \mathbf{1}(\|\varepsilon\| > r)] \\ &\leq \sum_{i=1}^D \sqrt{\mathbb{E}|\varepsilon_i|^4} \sqrt{\mathbb{P}(\|\varepsilon\| > r)} \\ &= D\sigma^2\sqrt{3} \sqrt{\mathbb{P}(\|\varepsilon\| > r)} \\ &\leq D\sigma^2\sqrt{3} \exp\left(-\frac{(r - \sigma\sqrt{D})^2}{4\sigma^2}\right) \end{aligned}$$

where the first inequality follows from the Cauchy–Schwarz inequality, the second equality follows from the fourth-moment formula for a centered Gaussian random variable, and the last inequality follows from (72). ■

Appendix B. Additional diagnostics for the latent-smoothing experiment

This appendix reports the complete Wasserstein curves for the parameter sweeps in Section 5.2; see Figures 8 to 11. Curves are means over five independent runs, with pointwise 95% Student- t confidence bands. Each panel also includes the direct ambient-smoothing reference $\hat{\mu}_n * \mathcal{N}(0, \sigma^2 I_D)$. Circular markers denote the bandwidth grid points with the smallest computed transport costs. The insets show

$$\Delta \widehat{W}_2^2(\sigma) := \widehat{W}_2^2(\sigma) - \widehat{W}_2^2(0)$$

on a symmetric logarithmic vertical scale, preserving the sign of the change while resolving small improvements near zero.

B.1 Reconstruction-error sensitivity

For the reconstruction-error sweep, we vary

$$\delta \in \{0, 0.2, 0.4, 0.6, 0.8\}$$

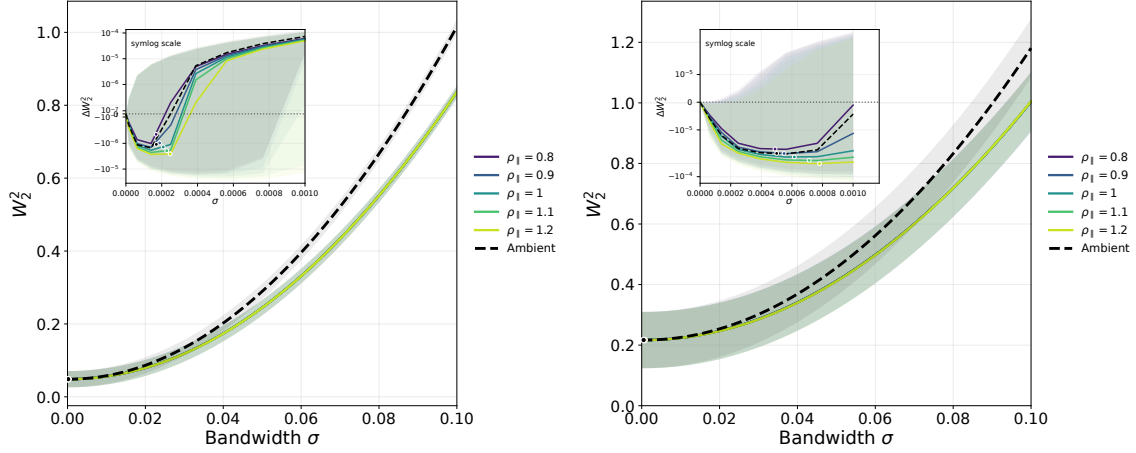


Figure 8: Complete Wasserstein curves for the tangential-gain sweep. Left: $\mathbb{S}^2$. Right: $\mathbb{S}^1 \times \mathbb{S}^2$.

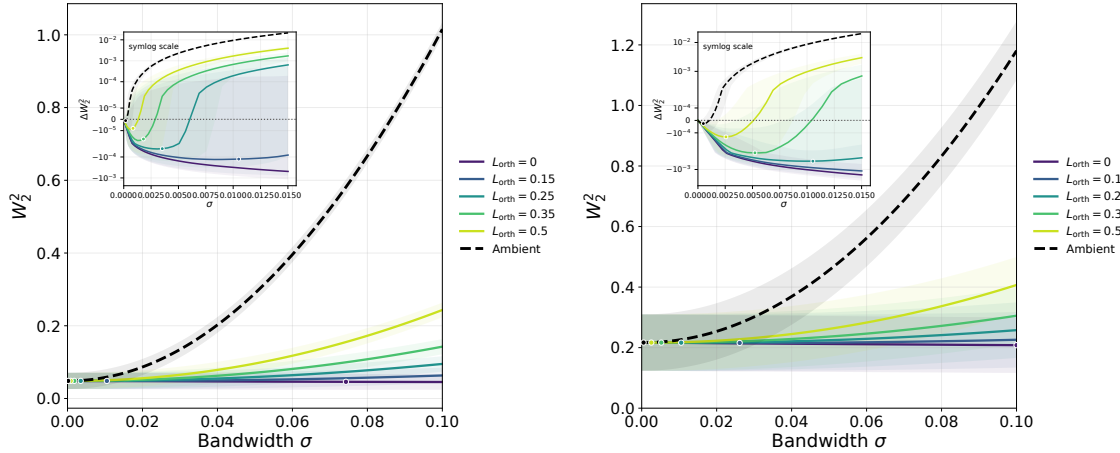


Figure 9: Complete Wasserstein curves for the orthogonal-response sweep. Left: $\mathbb{S}^2$. Right: $\mathbb{S}^1 \times \mathbb{S}^2$.

while fixing $\rho_{\parallel} = 1$, $L_{\text{orth}} = 1$, and $d - m = 80$. Changing δ translates every decoded output by the same vector δb without changing the decoder Jacobian. Among the four sweeps, this parameter produces the largest change in the vertical level of the Wasserstein curves. This agrees with the leading term $(W_0 + \delta)^2$ in Theorem 3: unlike the explicit orthogonal penalty, reconstruction error is already present at $\sigma = 0$.

We do not fit a universal law for $\sigma^*(\delta)$. The same parameter also enters the first- and second-order bandwidth-dependent coefficients in the theorem, so their competing effects do not imply a parameter-free monotone formula for the minimizer. Accordingly, the full curves in Figure 11 are presented as a sensitivity analysis. They show separately how reconstruction error changes the baseline discrepancy and whether a positive-bandwidth improvement remains.

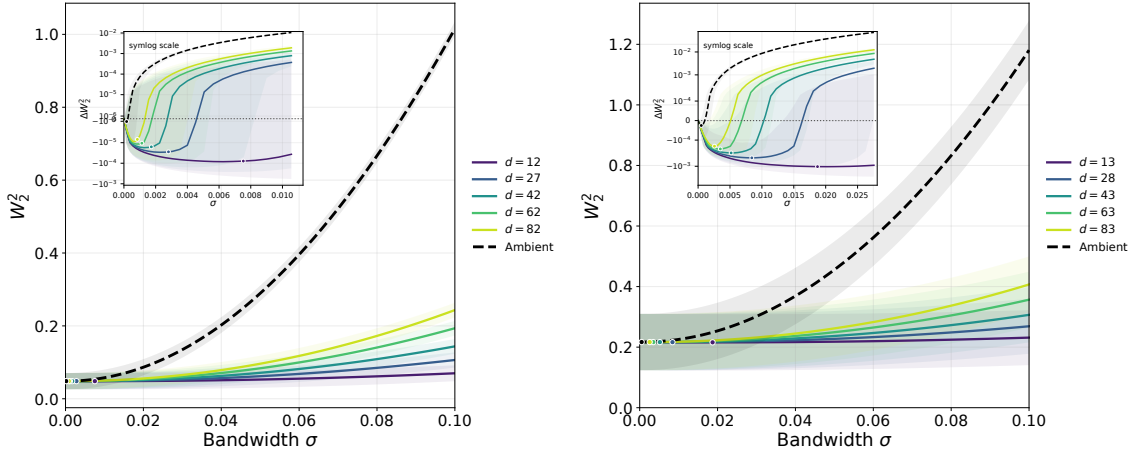


Figure 10: Complete Wasserstein curves for the latent-dimension sweep. Left: S^2 . Right: $S^1 \times S^2$.

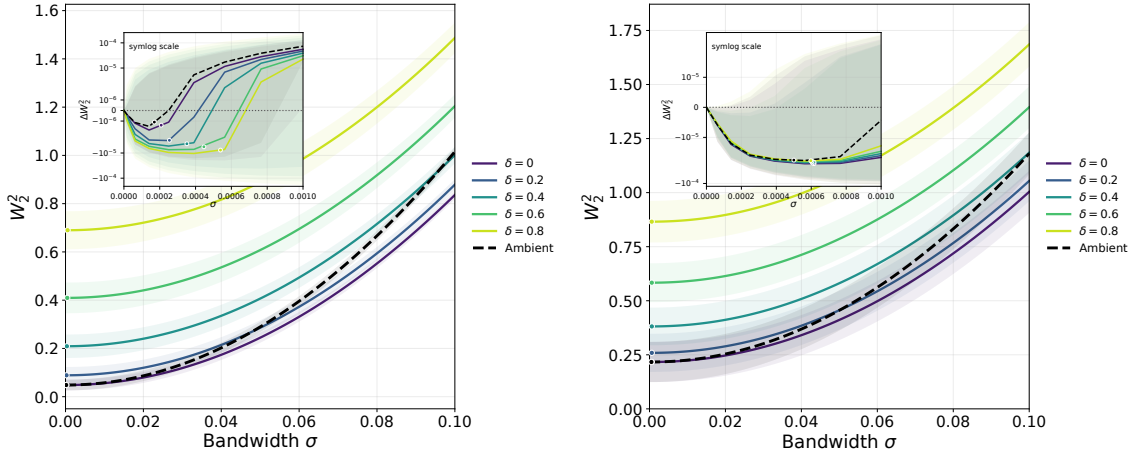


Figure 11: Sensitivity of the complete Wasserstein curves to reconstruction error. Left: S^2 . Right: $S^1 \times S^2$.

Appendix C. MNIST Implementation Details and Diagnostic Metrics

This section provides the architecture, training objectives, optimization settings, local-frame construction, and diagnostic formulas used in the MNIST study of Section 5.3. The synthetic experiments are used to evaluate the geometric quantities in settings with known tangent spaces. For MNIST, the local-frame calculations below are exploratory diagnostics of the trained network relative to data-estimated PCA directions.

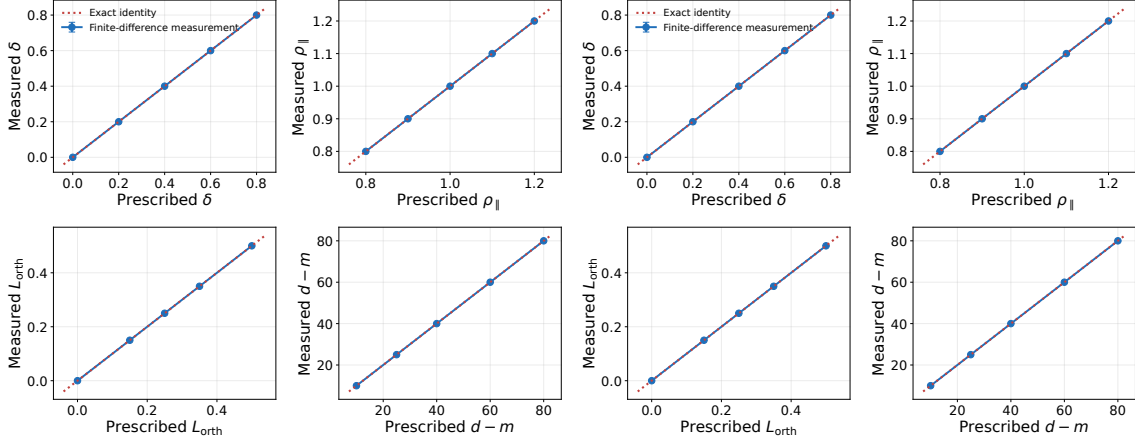


Figure 12: Numerical verification of the prescribed parameters in the explicit encoder-decoder construction. The left group shows the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^{100}$ with intrinsic dimension $m = 2$, and the right group shows the product manifold $\mathbb{S}^1 \times \mathbb{S}^2 \subset \mathbb{R}^{100}$ with intrinsic dimension $m = 3$. Within each group, the four panels compare the prescribed and measured values of the reconstruction error δ (top left), tangential gain $\rho_{\parallel}$ (top right), orthogonal response L_{orth} (bottom left), and latent-complement dimension $d - m$ (bottom right). Jacobian-based responses are evaluated by central finite differences at the empirical anchors. In every panel, the horizontal axis gives the prescribed value and the vertical axis gives its numerical measurement. The red dotted line is the identity $y = x$, representing exact agreement between the prescribed and measured quantities.

C.1 Network Architecture and Training Protocol

The Geometry-Preserving Encoder-Decoder (GPE) of [Lee et al. \(2025\)](#) is trained on the balanced empirical measure

$$\hat{\mu}_n = \frac{1}{20} \sum_{i=1}^{20} \delta_{x_i},$$

which contains two images from each MNIST digit class. The sample is selected using random seed 42. The encoder and decoder are trained sequentially for 2,000 epochs each, using full batches and the Adam optimizer with learning rate 10^{-3} .

Encoder training. The encoder $f_{\theta} : \mathbb{R}^{784} \rightarrow \mathbb{R}^{15}$ is a fully connected network with layer dimensions

$$784 \longrightarrow 512 \longrightarrow 256 \longrightarrow 128 \longrightarrow 15.$$

Softplus activations are applied after the three hidden layers. The encoder parameters are trained by minimizing

$$\mathcal{L}_{\text{enc}}(\theta) = \mathbb{E}_{x_i, x_j \sim \hat{\mu}_n} \left[\left(\log \frac{1 + \|f_{\theta}(x_i) - f_{\theta}(x_j)\|_2^2}{1 + \|x_i - x_j\|_2^2} \right)^2 \right]. \quad (73)$$

This objective penalizes discrepancies between pairwise distances of the training anchors and their latent representations. Because it is evaluated on a finite set of anchors, it does

not by itself determine the Jacobians $J_f(x_i)$ or verify local isometry on an unknown MNIST tangent space.

Decoder training. After freezing the encoder, the decoder $g_\phi : \mathbb{R}^{15} \rightarrow [0, 1]^{784}$ is trained. Its layer dimensions are

$$15 \longrightarrow 128 \longrightarrow 256 \longrightarrow 512 \longrightarrow 784.$$

The three hidden layers use Softplus activations, and the output layer uses a sigmoid activation. At every training iteration, independent noise

$$\xi \sim \mathcal{N}(0, \sigma_{\text{train}}^2 I_{15}), \quad \sigma_{\text{train}} = 0.05,$$

is added to the latent representation. The decoder objective is

$$\mathcal{L}_{\text{dec}}(\phi) = \mathbb{E}_{x \sim \hat{\mu}_n, \xi} [\|g_\phi(f_\theta(x) + \xi) - x\|_2^2]. \quad (74)$$

The implementation averages the squared error over pixels, which differs from (74) only by the fixed factor 784 and therefore has the same minimizers.

C.2 Reconstruction and Local-Frame Diagnostics

MNIST does not provide a known intrinsic dimension or ground-truth tangent spaces. We therefore do not report an empirical encoder-isometry defect. The quantities defined below measure the network response relative to specified local PCA frames and are not used as estimates of the intrinsic constants in Theorem 3.

Local PCA frames. For each training anchor x_i , we construct an orthonormal matrix

$$E_i \in \mathbb{R}^{784 \times m}$$

by anchor-centered, distance-weighted PCA using same-class neighbors from the full MNIST training collection. The offset $x_j - x_i$ receives the weight

$$w_{ij} = \exp \left[-\frac{1}{2} \left(\frac{\|x_j - x_i\|_2}{h_i} \right)^2 \right].$$

The primary configuration uses $m = 10$, $k = 30$ neighbors, and

$$h_i = 0.75 \text{ median}_{j \in \mathcal{N}_i} \|x_j - x_i\|_2.$$

As a sensitivity check, we also use $k = 50$ neighbors with

$$h_i = \text{median}_{j \in \mathcal{N}_i} \|x_j - x_i\|_2.$$

The larger MNIST collection is used only after training, for this local-frame construction and for the finite reference measure in the transport comparison.

Reconstruction displacement. The empirical RMS Euclidean reconstruction displacement is

$$\hat{\delta} := \left(\frac{1}{n} \sum_{i=1}^n \|x_i - g_\phi(f_\theta(x_i))\|_2^2 \right)^{1/2}. \quad (75)$$

This Euclidean displacement should be distinguished from the pixel-averaged mean squared error used during decoder training.

Encoded local frame. For each estimated frame, define $B_i := J_f(x_i)E_i \in \mathbb{R}^{15 \times m}$. We record the numerical rank and extreme singular values of B_i to determine whether the selected m -dimensional frame remains m -dimensional after encoding. These values are numerical checks on the selected PCA frames, not estimates of an intrinsic encoder-isometry constant.

Shared PCA-frame tangential gain and residual. The composed Jacobian acting on the estimated local frame is

$$A_i := J_g(f_\theta(x_i))B_i = J_g(f_\theta(x_i))J_f(x_i)E_i \in \mathbb{R}^{784 \times m}.$$

Relative to these frames, we define the shared tangential gain and its operator-norm residual by

$$\hat{\rho}_{\parallel} \in \arg \min_{\rho \geq 0} \max_{1 \leq i \leq n} \min_{O_i \in O(m)} \|A_i - \rho E_i O_i\|_{\text{op}}, \quad (76)$$

$$\hat{\varepsilon}_{\text{iso}} := \max_{1 \leq i \leq n} \min_{O_i \in O(m)} \|A_i - \hat{\rho}_{\parallel} E_i O_i\|_{\text{op}}. \quad (77)$$

Both quantities remain conditional on the chosen PCA frames.

For the numerical optimization, write $E_i^\top A_i = U_i \Sigma_i W_i^\top$ and initialize the orthogonal factor by $O_{i,0} = U_i W_i^\top$. This is the orthogonal Procrustes solution for the Frobenius norm and is used only as an initialization for the operator-norm objective. At fixed ρ , we refine O_i through the parameterization

$$O_i = O_{i,0} \exp(K_i - K_i^\top).$$

Both connected components of $O(m)$ are considered. The refinement uses 300 Adam iterations with learning rate 3×10^{-2} , together with one random orthogonal restart. We alternate this refinement with a bounded one-dimensional minimization over the shared variable ρ for at most four iterations. Final operator norms are evaluated by singular value decomposition rather than by the smooth objective used during refinement.

As an independent numerical check, we approximate $J_{g \circ f}(x_i)E_i$ by centered finite differences with step size 10^{-2} and compare the result with the automatic-differentiation Jacobian.

Latent-complement decoder response. Let $r_i := \text{rank}(B_i)$, let $Q_i \in \mathbb{R}^{15 \times r_i}$ have orthonormal columns spanning $\text{range}(B_i)$, and let $Q_i^\perp \in \mathbb{R}^{15 \times (15-r_i)}$ span its orthogonal complement. We define the PCA-frame latent-complement response by

$$\hat{L}_{\text{orth}} := \max_{1 \leq i \leq n} \left\| J_g(f_\theta(x_i))Q_i^\perp \right\|_{\text{op}}. \quad (78)$$

The complement is constructed from $\text{range}(J_f(x_i)E_i)$ rather than from the range of the full encoder Jacobian. If $r_i = 15$, the complement is empty and its contribution is zero by convention.

Table 1: Empirical diagnostics for the GPE trained on $\hat{\mu}_n$, with $n = 20$, $d = 15$, and $D = 784$. The local PCA calculations use the estimated dimension $m = 10$ and the primary $k = 30$ neighborhood configuration. The reconstruction displacement is computed directly at the training anchors. The remaining quantities are conditional on the data-estimated PCA frames and are not interpreted as intrinsic geometric constants in Theorem 3.

Metric	Symbol	Empirical value
RMS reconstruction displacement	$\hat{\delta}$	1.87×10^{-3}
PCA-frame tangential gain	$\hat{\rho}_{\parallel}$	1.98×10^{-3}
PCA-frame tangential residual	$\hat{\epsilon}_{\text{iso}}$	4.34×10^{-3}
Latent-complement decoder response	$\hat{L}_{\text{orth}}$	3.83×10^{-3}

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